



Republic of Tunisia
Ministry of Higher Education
and Scientific Research
University of Tunis El Manar
Faculty of Sciences Tunis, El Manar
Information Technology Department



End of Studies Project

Web & mobile apps

Promoting physical activity & low-carbon mobility

Realised by : Siwar Soula

2023-2024



Academic supervisor :

Mrs Ines Riahi

Professional supervisor :

Mr Injez Temimi



Linedata

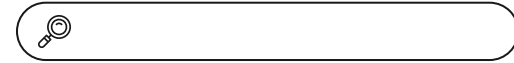


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01

General Introduction



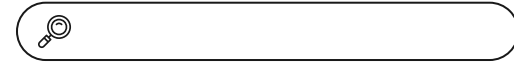


Hosting Organism

Linedata Technologies Of Tunisia

- Founded in 1998
- Global provider of **software**, services and value-added data dedicated to the **financial** sector
- Committed to integrating **CSR** principles into its business model





Project context



Corporate social responsibility

CSR is the idea that businesses should integrate social and environmental concerns in their business operations.

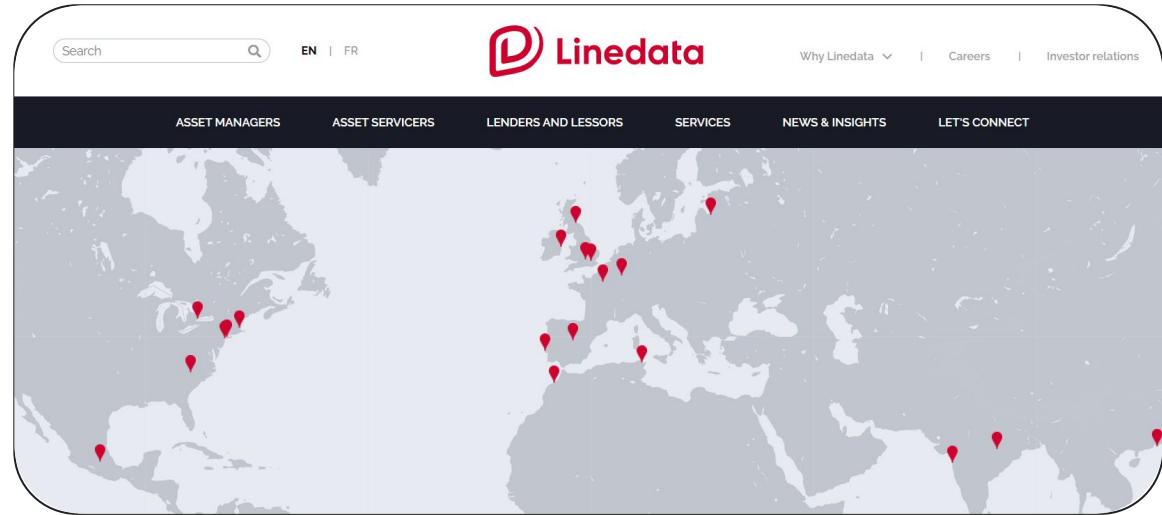


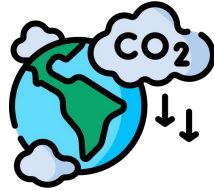
20

Offices

1100

employees

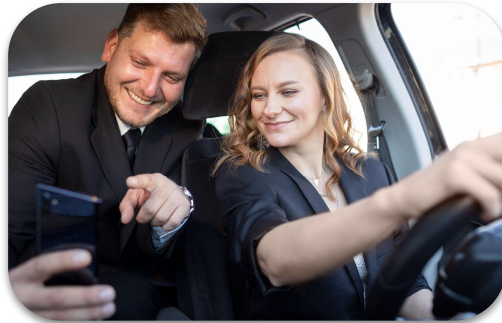




4,400,000 CO₂

kilograms

**the average person produces about 4 tons of
carbon dioxide each year**

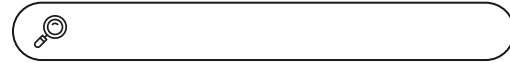


Green mobility
is the movement of people in
any transportation mode with
a low environmental effect.



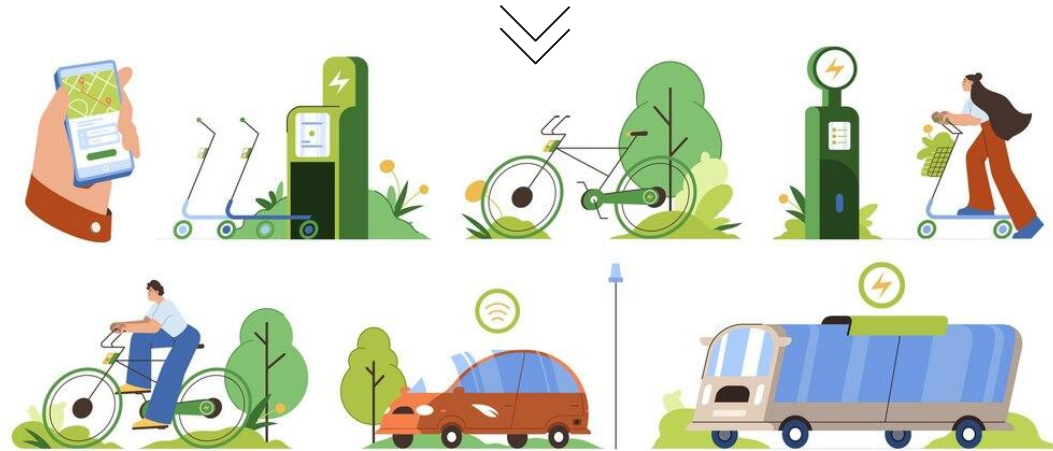
Reduce Linedata's employees' carbon footprint through an engaging and user-friendly mobile application.

Project Objective



Proposed Solution

GREEN MOBILITY CHALLENGE





Mobile Application

Progress Tracking

Monitor actions



Incentive System

Earning points



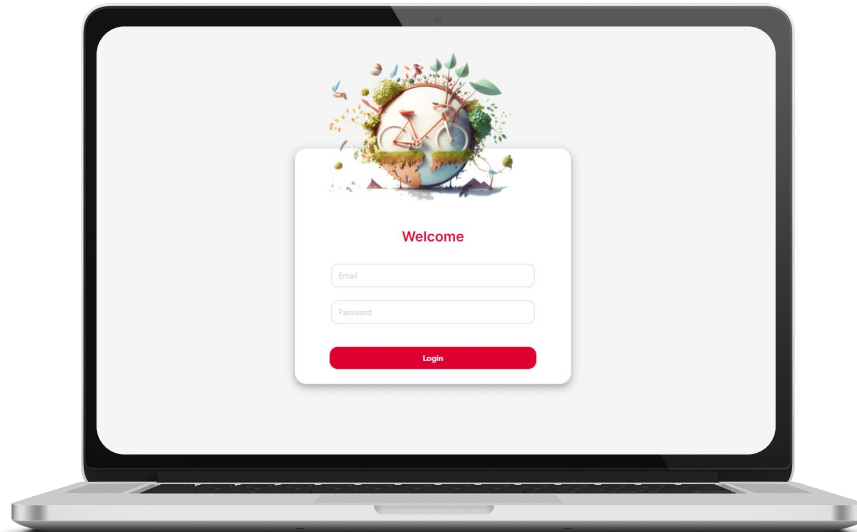
Community Engagement

Teamwork & Information





Web Application



Events Management

Results, activities & functionalities, quizzes & surveys



Users

Roles & synchronization

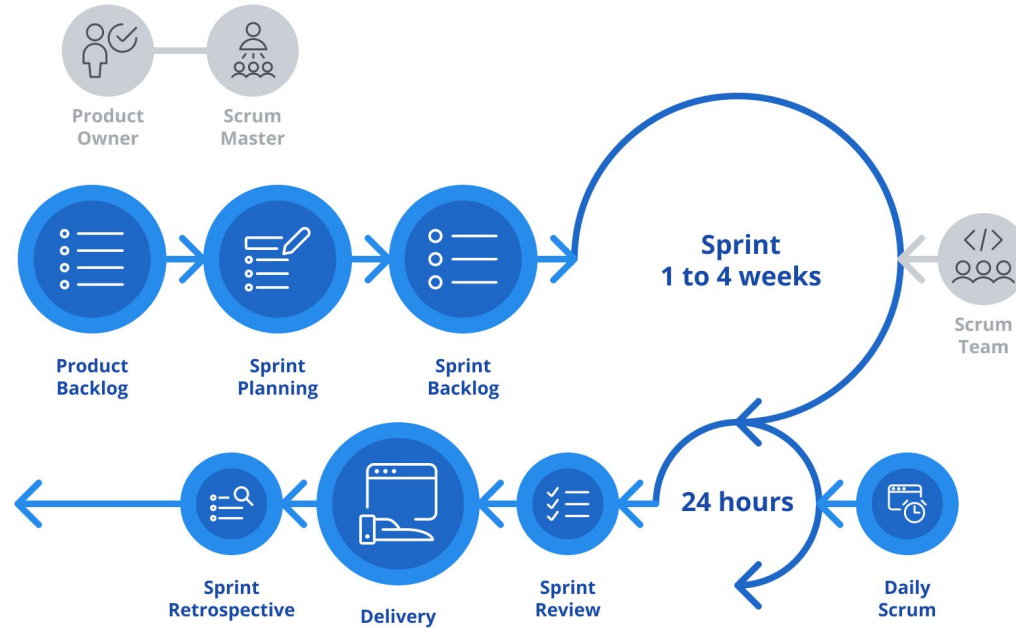


Community Engagement

Articles & Posts & Notifications



Methodology





Admin



User

story mapping

Account

Navigation

Manage Daily Progress

Manage events

Manage Teams

Sign in

Logout

View profile

Edit profile

general visual

App navigation

View daily general stats

View all events

Preview one Event

Participate at an event

Create team

Edit team

Sign in with username & password

Logout after sign in

View profile

Edit user details

Common layout

Routing & Navigation

view general personal progress

view dashboard as homepage

preview an event from events table

view event tasks

view event personal progress

create team

edit team details

manage team members

Create Login Form

on logout redirect user to Login

Create profile card

Update user password funct

general header

Start on dashboard

Create Daily progress Card

Create events table

Create Event preview Card

Create Event progress Card

Create Event preview Card

change team image or name

manage team join request

authenticate user

user details data binding

save new user details

general footer

Create all pages

event details data binding

events details data binding

one event details data binding

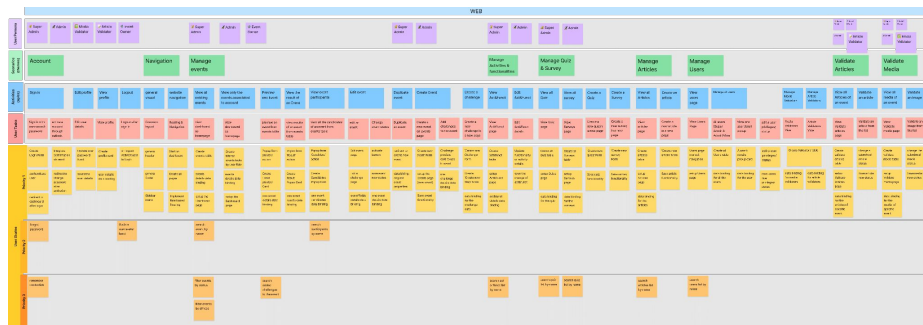
event details data binding

delete members

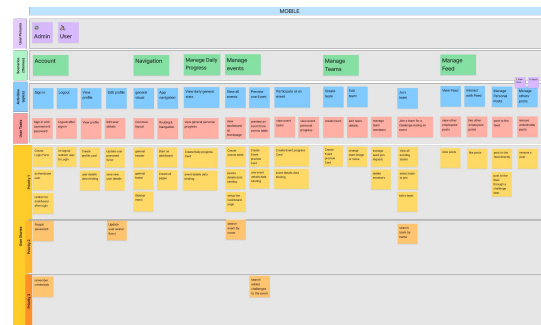
redirect to dashboard after login

Sidebar menu

setup the Dashboard page



Web application



Mobile application

story mapping

Linedata Backlog Sprint My Tasks Administration Global Viewer Epic Resource Manager KEN Challenge Vert Project

second_sprint Progress 75% Total Distractions: 0 (Hours)

To Do 0	In Progress 0	In Development 4	QA 0	Done 14
1 ST0000099786 - design de l'application (frontend)				
To Do 0	In Progress 0	In Development 0	QA 0	Done 1
2 ST0000099785 - rédaction cahier de charge				
To Do 0	In Progress 0	In Development 0	QA 0	Done 2
3 ST0000099784 - définir la conception				
To Do 0	In Progress 0	In Development 0	QA 0	Done 2
4 ST0000099820 - CRUD Functionality Back				
To Do 0	In Progress 0	In Development 0	QA 0	Done 1
5 ST0000099832 - CRUD Functionality Front				
To Do 0	In Progress 0	In Development 0	QA 0	Done 1
6 ST0000099821 - CRUD Activity Back				
To Do 0	In Progress 0	In Development 0	QA 0	Done 1
7 ST0000099822 - CRUD Survey Back				
To Do 0	In Progress 0	In Development 1	QA 0	Done 0
8 ST0000099823 - CRUD Quizz Back				
To Do 0	In Progress 0	In Development 1	QA 0	Done 0
9 ST0000099827 - Update Event Back				
To Do 0	In Progress 0	In Development 0	QA 0	Done 0
10 ST0000099828 - Events Dashboard Back				
To Do 0	In Progress 0	In Development 0	QA 0	Done 1
11 ST0000099833 - CRUD Activity Front				
To Do 0	In Progress 0	In Development 0	QA 0	Done 0
12 ST0000099838 - CRUD Survey Front				
		CRUD Survey Front		

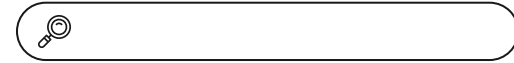
Sprint Planning



02

Analysis & Requirements Specification





Requirements Specification



Walk



Bike



Public Transport



Carpooling



Walk after Public
Transport



Quiz



Survey



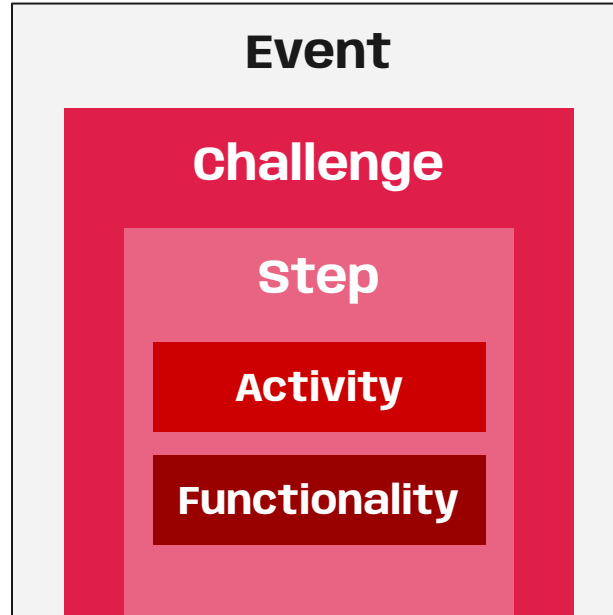
Activities

Functionalities



Requirements Specification

Event "Green
Mobility Challenge"
Specification



- Walk
- Bike
- Public Transport
- Carpooling
- Walk after Public Transport
- Quiz
- Survey



Requirements Specification

System Actors

- User



Event "Green
Mobility Challenge"
Specification



Walk



Bike



Public Transport



Carpooling



Walk after Public
Transport



Quiz



Survey



Requirements Specification

System Actors

- User
- Owner
- Admin
- Super Admin



Event "Green Mobility Challenge" Specification



Walk



Bike



Public Transport



Carpooling



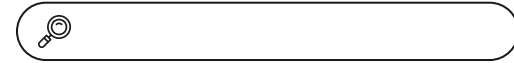
Walk after Public Transport



Quiz



Survey



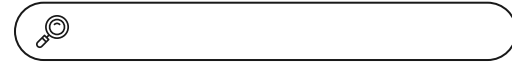
Functional Requirements



- Authentication
- Manage Account



- Manage Events
- Manage Activities & Functionalities
- Manage Quizzes & surveys
- Manage Users
- Manage Feed Posts (Media)
- Manage Notifications
- Manage Results
- Participate in an event
- Manage Daily Progress
- Manage Team
- Like posts



Non-functional requirements



Performance



Security



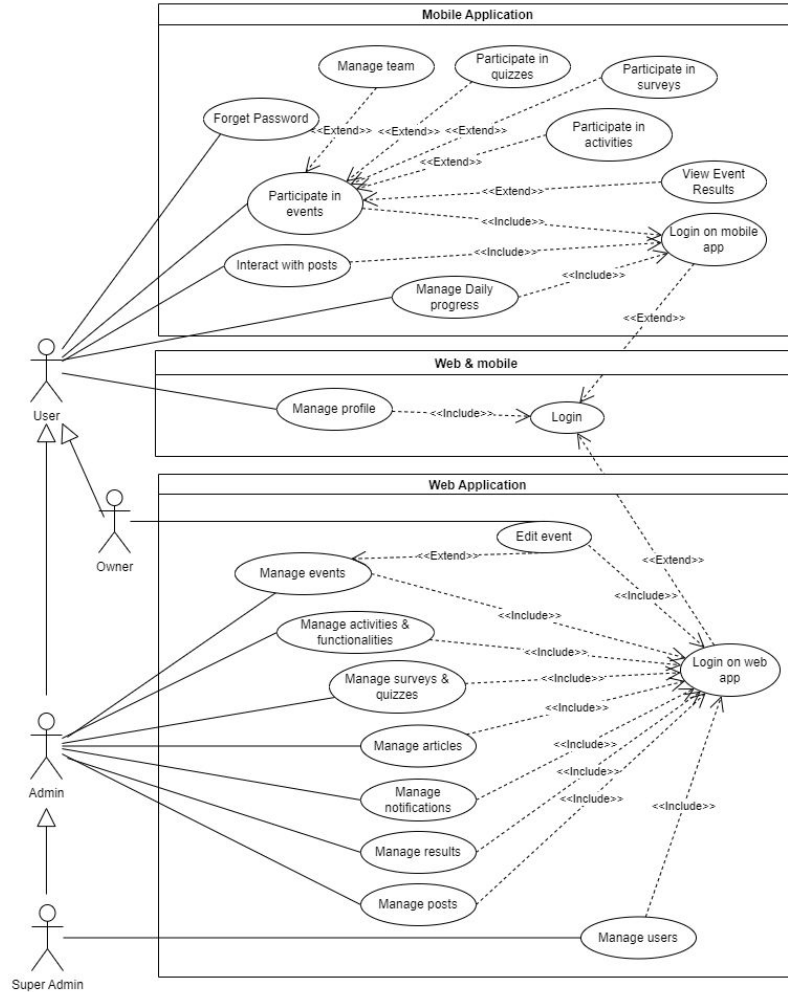
Usability

Availability



Scalability





Report page 17



Global Use Case Diagram



Requirements Specification

Core Features

Authentication

Events Management

Event participation

Team Management

Additional Features

Notifications management

Articles management

Posts management

Feed management



Release Planning

GANTT CHART

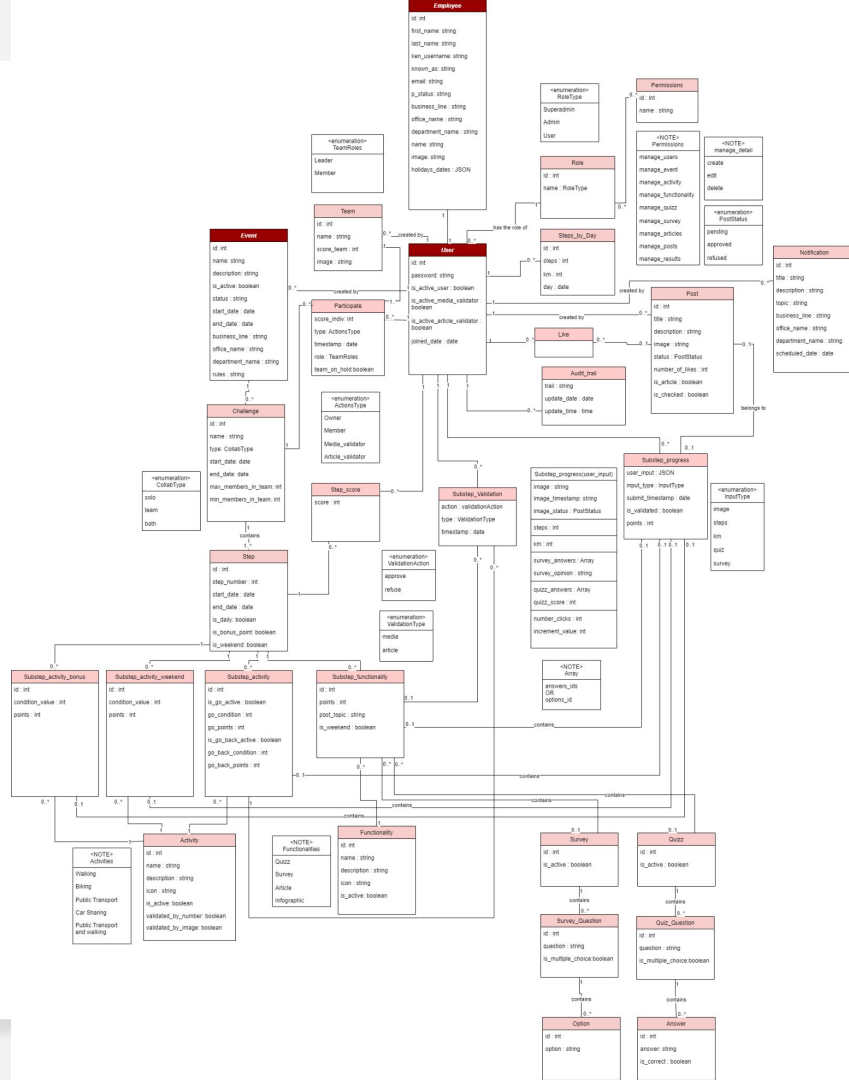


RELEASE PLANNING

1st release	2nd release	3rd release
PB-004 PB-009 Manage events	PB-001 PB-002 Web Account	PB-017 Manage Notifications
PB-003 Manage Activities & functionalities	PB-011 Participate in event	PB-008 Manage Articles
PB-006 Manage Quiz & Survey	PB-012 Manage Daily Progress	PB-007 Manage Media
PB-005 Manage Users	PB-013 PB-014 Manage Teams	PB-015 Manage Feed
	PB-010 PB-016 PB-019 Mobile Account	PB-018 Manage Results

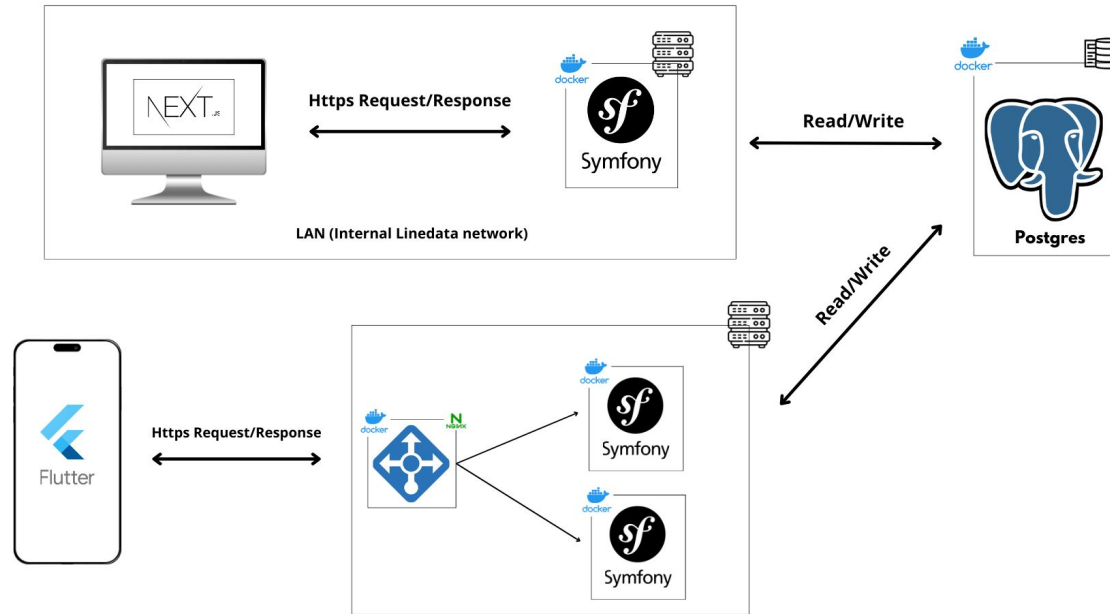
web app mobile app

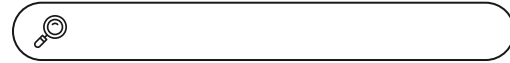
Report page 21



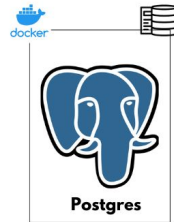


Architecture





Technical Choices & Tools





03

Sprint 1 : Web App Core Features





Sprint 1

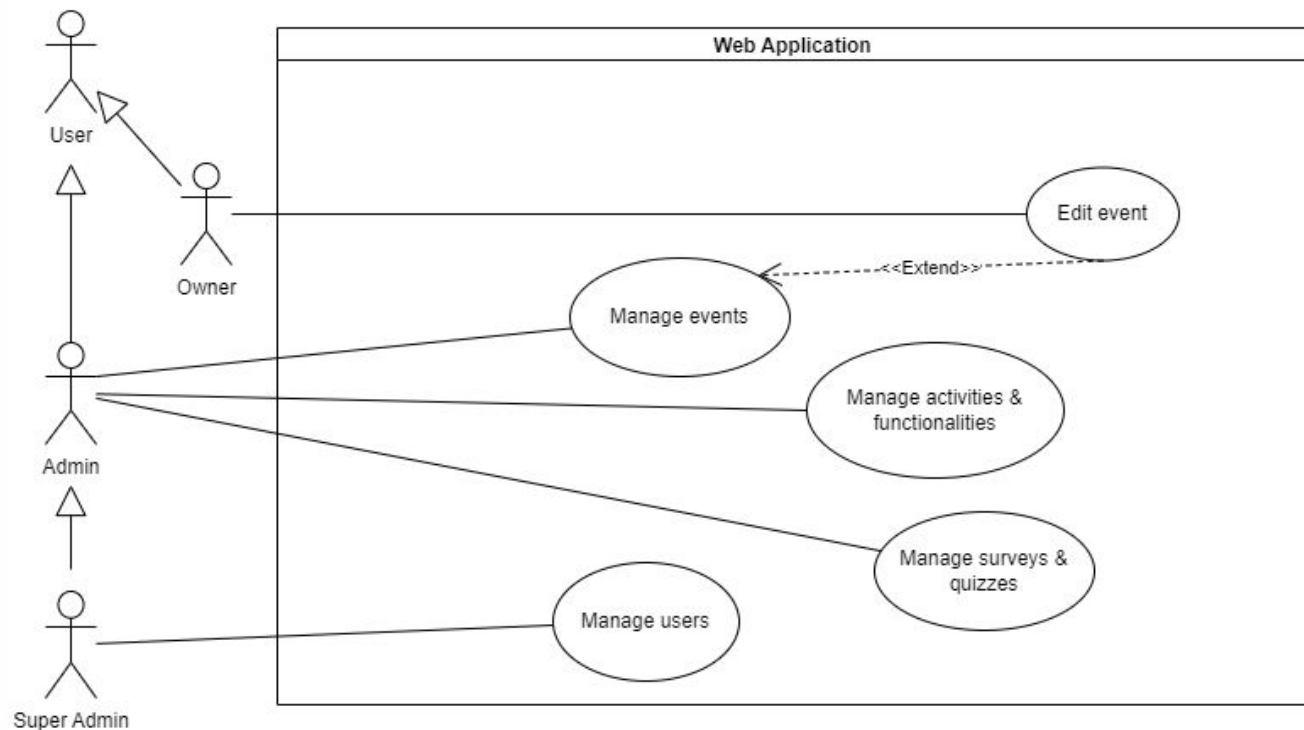


Use case diagram

p28



1st Sprint : Green Challenge Application





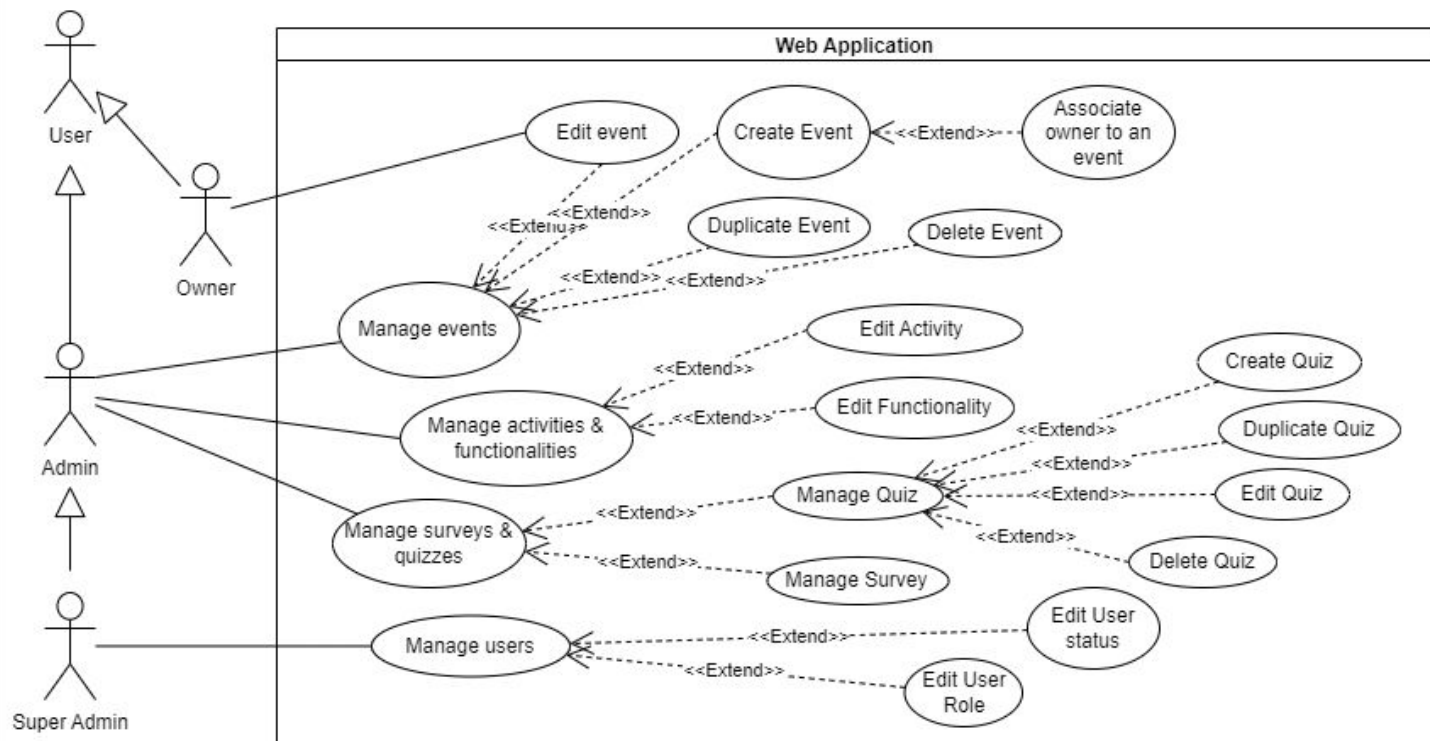
Sprint 1



Use case diagram detail



1st Sprint : Green Challenge Application





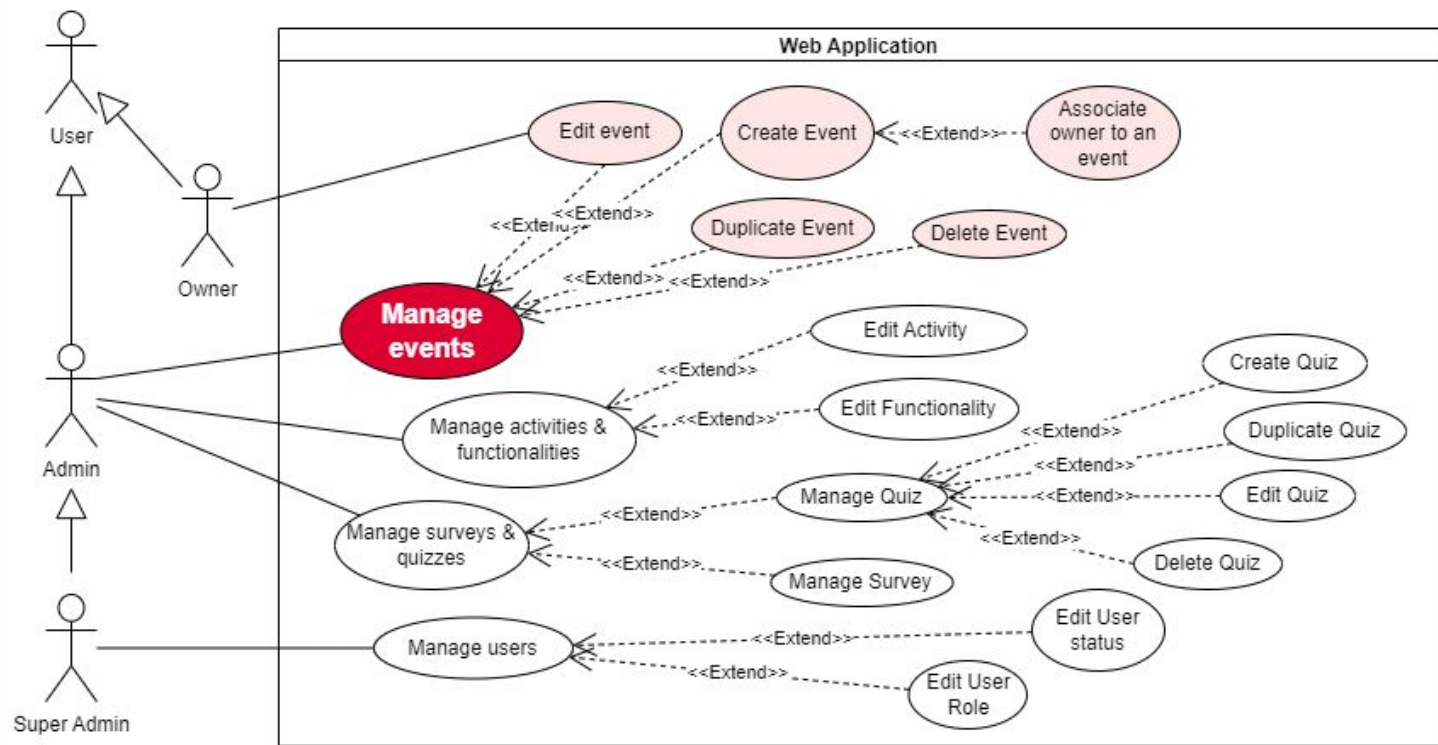
Sprint 1



Use case diagram detail



1st Sprint : Green Challenge Application





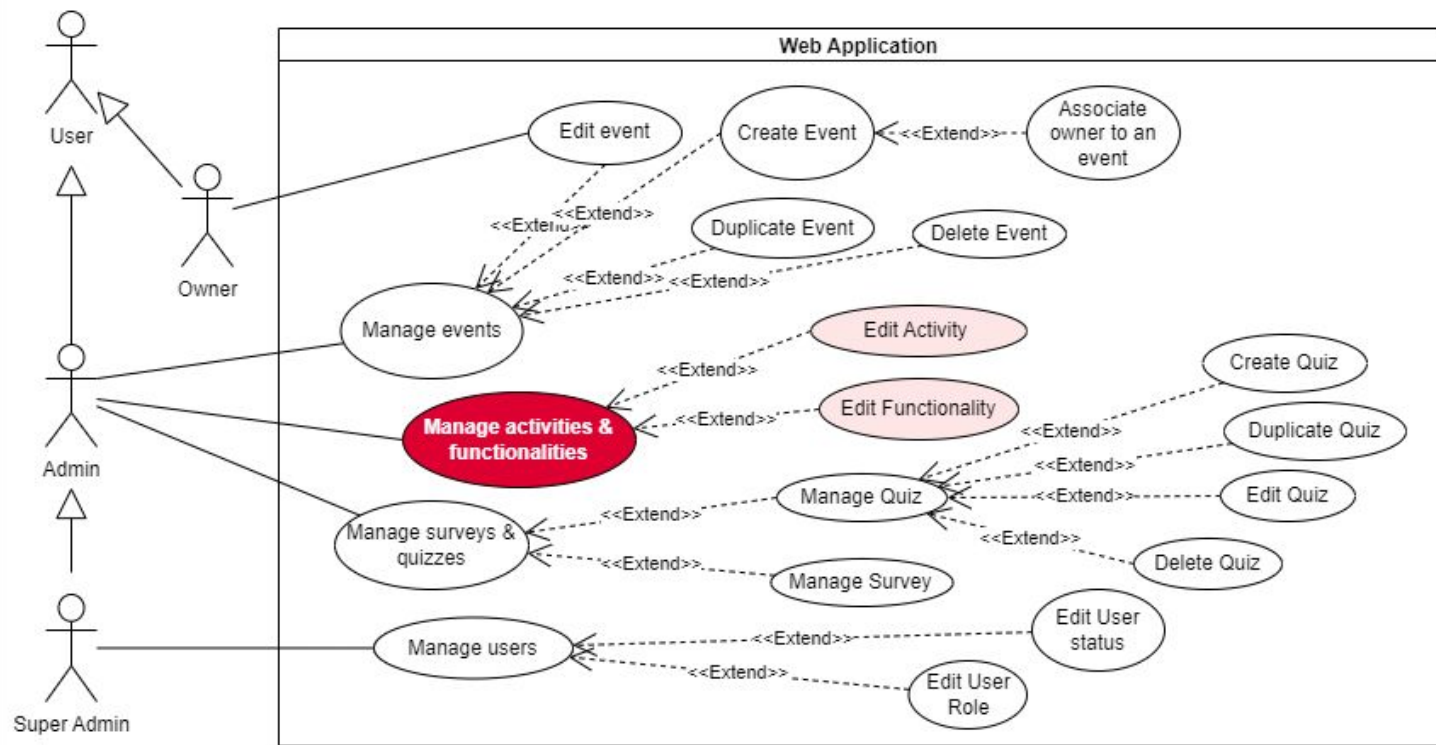
Sprint 1



Use case diagram detail



1st Sprint : Green Challenge Application





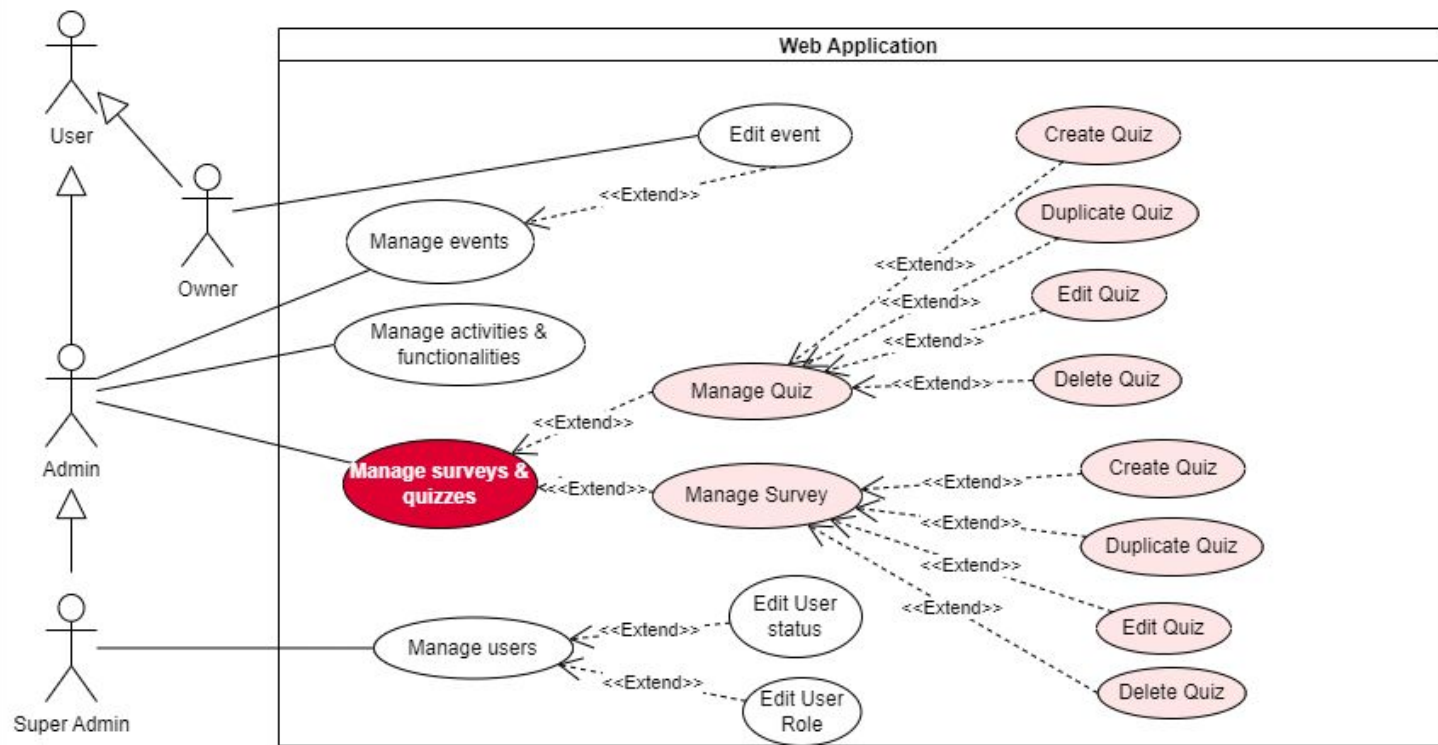
Sprint 1



Use case diagram detail



1st Sprint : Green Challenge Application





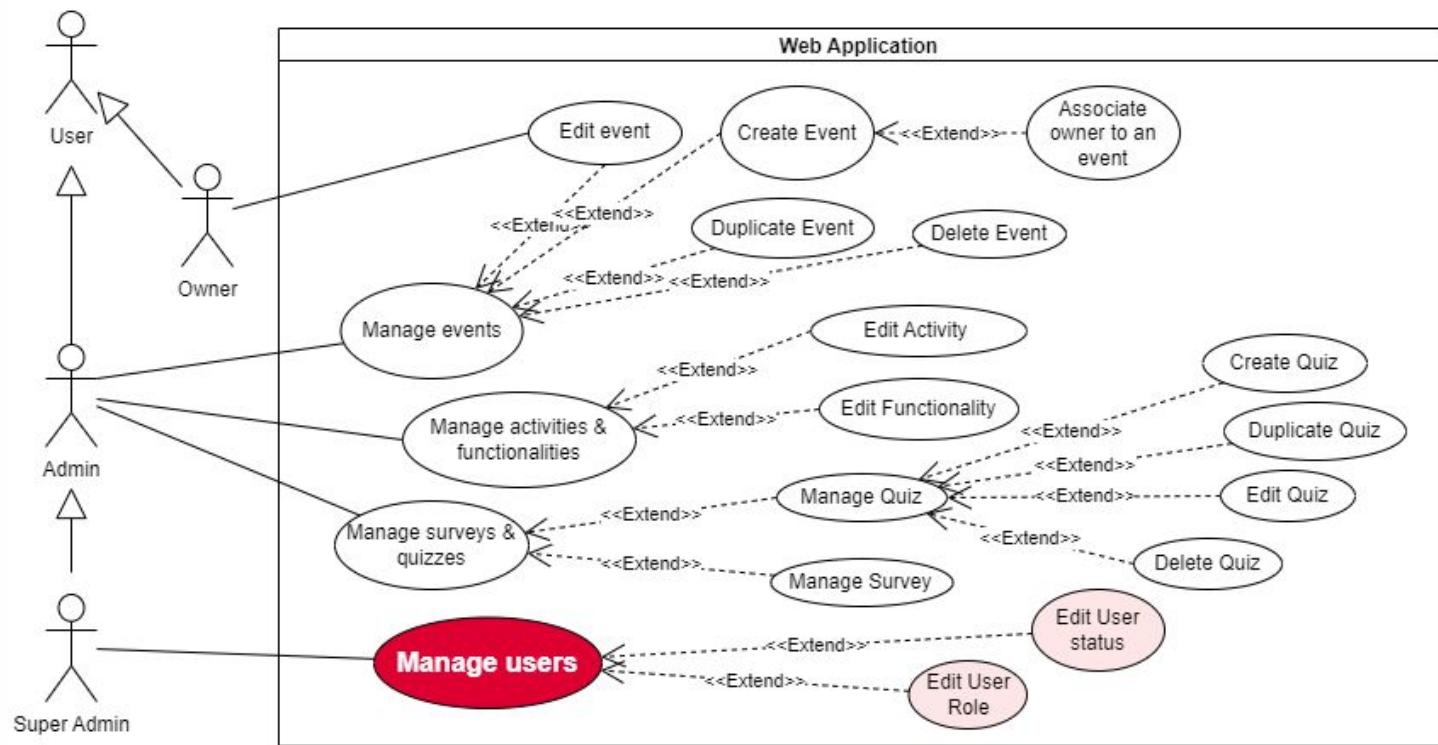
Sprint 1



Use case diagram detail



1st Sprint : Green Challenge Application



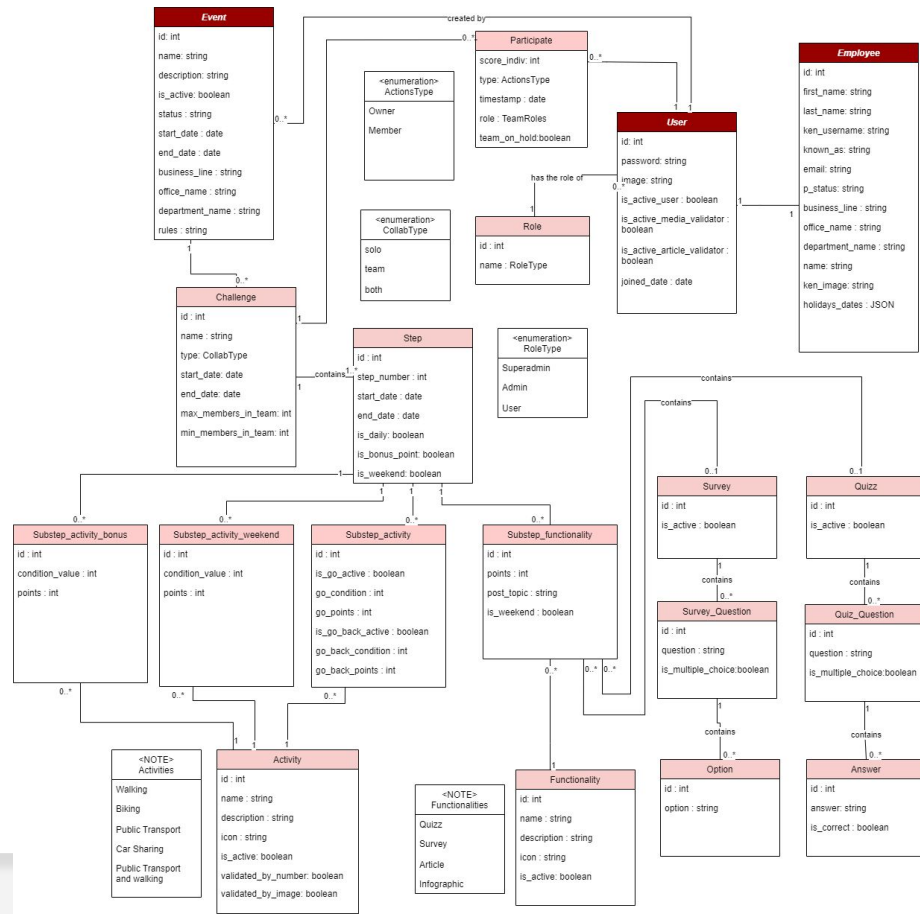


Sprint 1



Class Diagram

p38

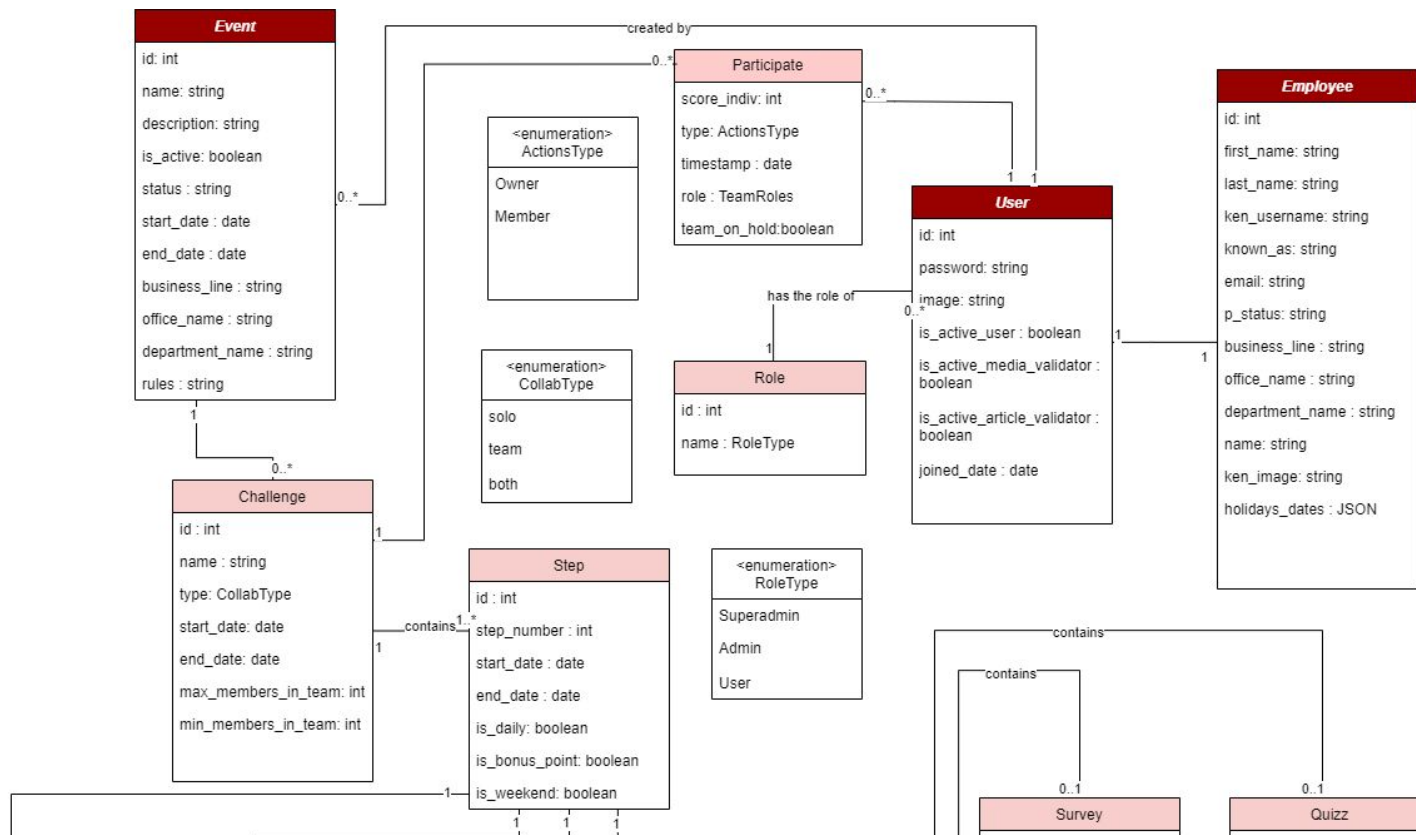




Sprint 1



Class Diagram

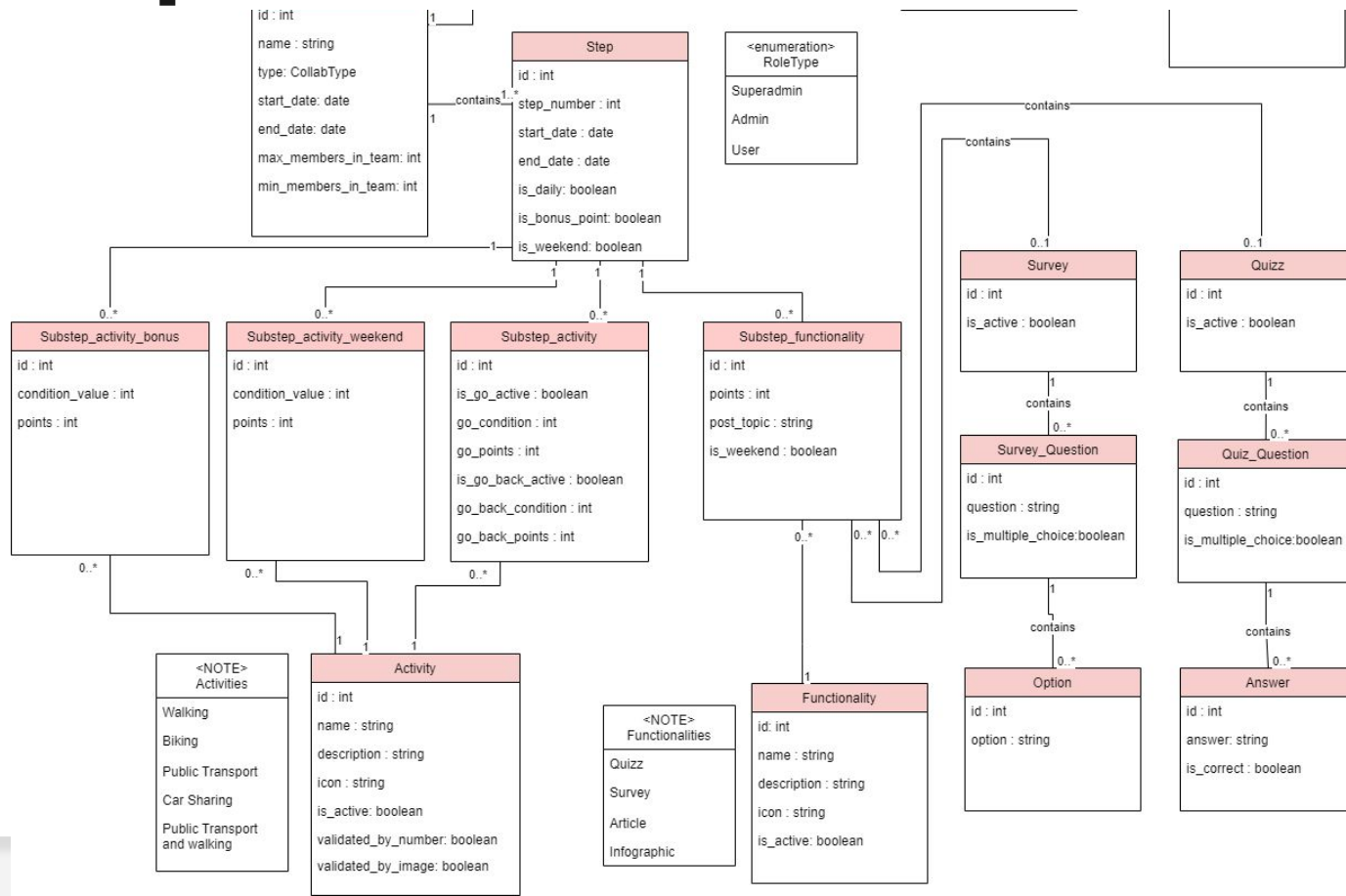




Sprint 1



Class Diagram



Create Event

Not secure In1challenge01.ad.linedata.com:2024/events/newEvent

Linedata Green Challenge

Dashboard
Activities & Functions
Quiz
Survey
Articles Post
Notifications
Users
Media Validation
Articles Validation

Create Event Status: Draft

Information Global

*Name: *Start: *End:

Description: Office: Business Line:

Group: Owner: Media Validators:

Article Validators:

H1 H2 H3 | **B** *I* U

Rules:

Challenges

☐ No challenges added !

Create Event

Add
Challenge

Not secure In1challenge01.ad.linedata.com:2024/events/newEvent

Linedata Green Challenge

Dashboard
Activities & Funct...
Quiz
Survey
Articles Post
Notifications
Users
Media Validation
Articles Validation

Create Event Status: Draft

Information Global

*Name: XXX *Start: 2024-08-17 *End: 2024-08-27 11 days

Description: xxxxx Office: Northern Euro... Asia Business Line: Gravitas Front Office

Group: FS: R&D - TA - ... TM: Corporate... Owner: Abdi Hersi Media Validators: Abdi Hersi Abhay Joshi

Article Validators: Aasim Nakhwa Abdi Hersi

H1 H2 H3 B I U

title
subtitle

Rules: more text

Add challenge

Challenges

No challenges added !

Create Challenge

Sub-step Activity

- Dashboard
- Activities & Function...
- Quiz
- Survey
- Articles Post
- Notifications
- Users
- Media Validation
- Articles Validation

Create Challenge

Information Global

*Name: xxxxx *Start: 2024-08-17 *End: 2024-08-27 11 days

☐ Solo ☒ Team ☐ Both 10 Max 2 Min Members in a team

Steps - 2 +

Step 1:

*Start: 2024-08-17 *End: 2024-08-20 4 days ☒ Daily ☒ Bonus Point ☒ Weekend

Activity ☒ ☒ ☐ ☐ ☐

Condition

☒ ☐ 2000 Steps 50 Points ☒ Go ☐ Go Back

☒ Upload media ☐ Points ☐ Go ☐ 50 Points ☒ Go Back

★ Weekend

☒ 5000 Steps 50 Points ☐
☒ 3 KM 200 Points ☐

Functionality ☐ ☒ ☒

Condition

☐ Choose survey satisfaction des c... 0 Points

☒ Choose quizz La sécheresse 0 Points

Bonus Point

☒ 5000 Steps 70 Points

Step 2 :

Submit Event

Dashboard

Activities & Function...

Quiz

Survey

Articles Post

Notifications

Users

Media Validation

Articles Validation

Create Event

Status: Draft

Information Global

*Name: XXX

*Start: 2024-08-17

*End: 2024-08-27

11 days

Description: xxxxxx

Office: Northern Euro...
Asia

Business Line: Gravitas
Front Office

Group: FS: R&D - TA - ...
TM: Corporate...

Owner: Abdi Hersi

Media Validators: Abdi Hersi
Abhay Joshi

Article Validators: Aasim Nakhwa
Abdi Hersi

H1 H2 H3 B I U

title

subtitle

Rules: more text

Add challenge

Challenges

XXXXX draft

team Start : 2024-08-17 End : 2024-08-27 11 days

daily Step 1 :

Activities :  

Functionalities :  

Weekend :  

Bonus : 

Cancel

Finish

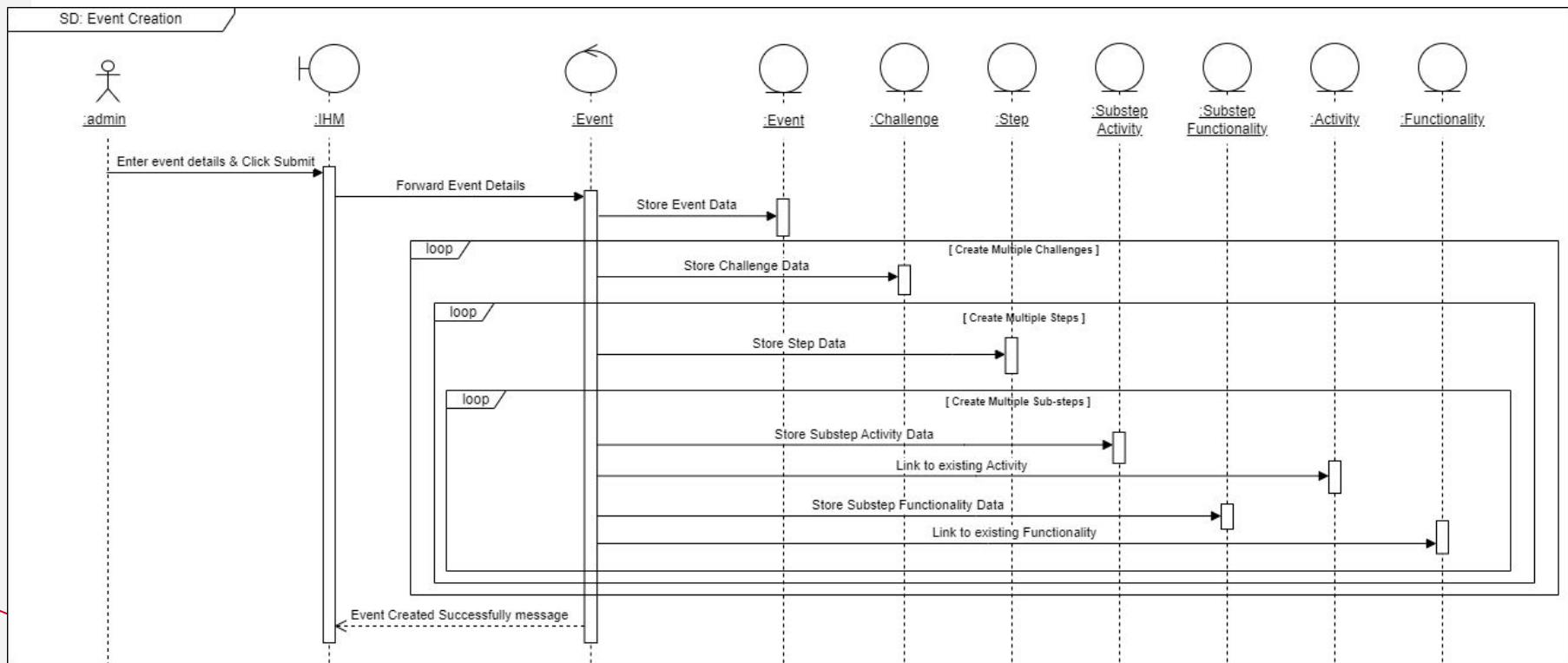
43



Create Event



Sequence Diagram p35





04

Sprint 2 : Mobile App Core Features





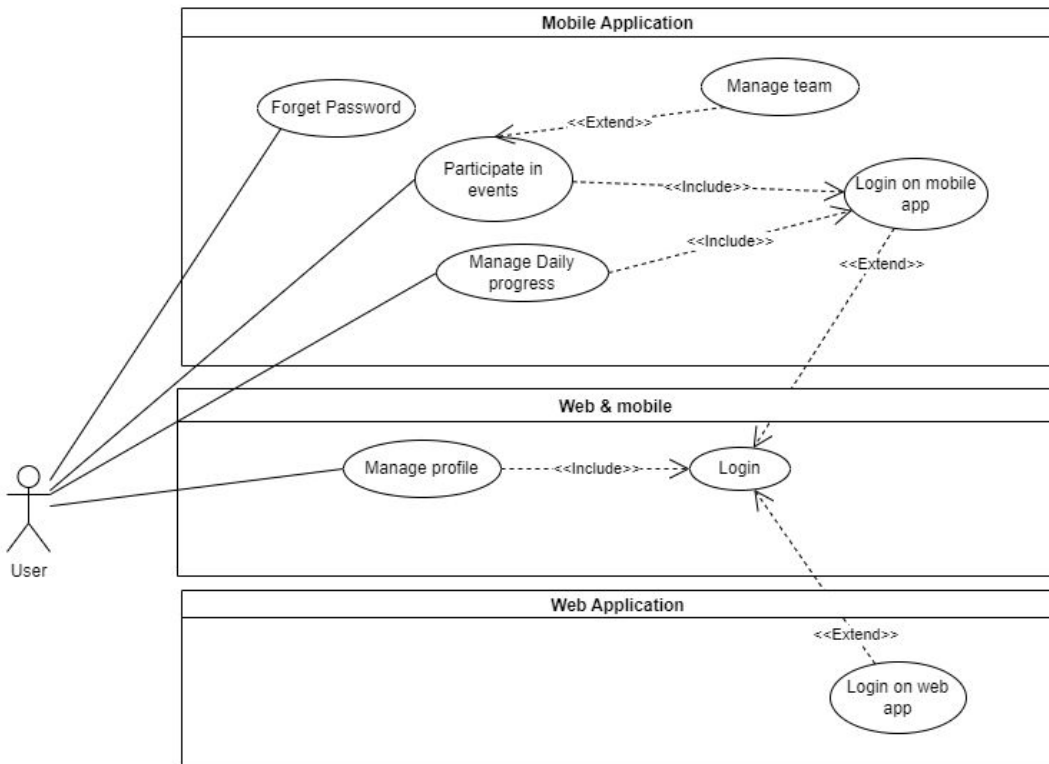
Sprint 2



Use case diagram p50



2nd Sprint : Green Challenge Application





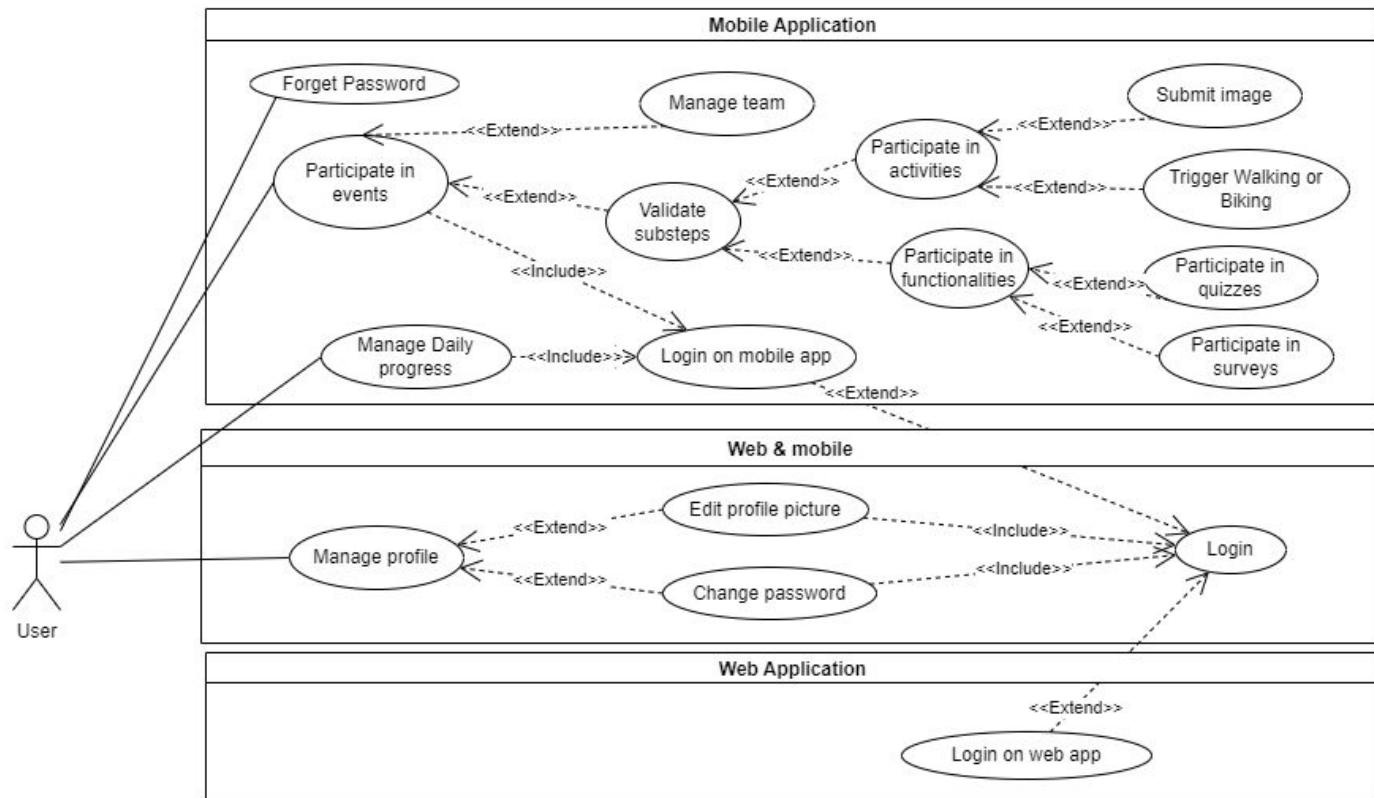
Sprint 2



Use case diagram detail



2nd Sprint : Green Challenge Application





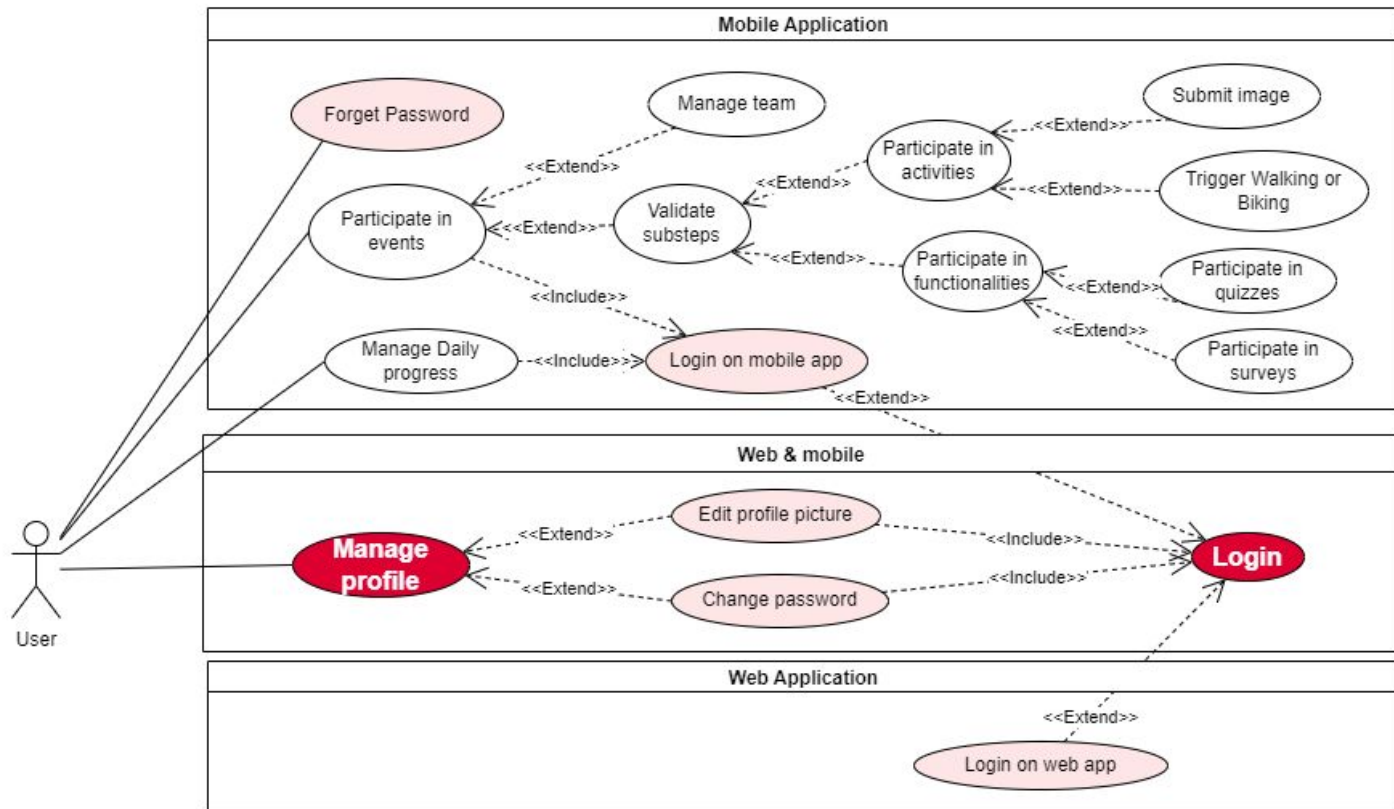
Sprint 2



Use case diagram detail



2nd Sprint : Green Challenge Application





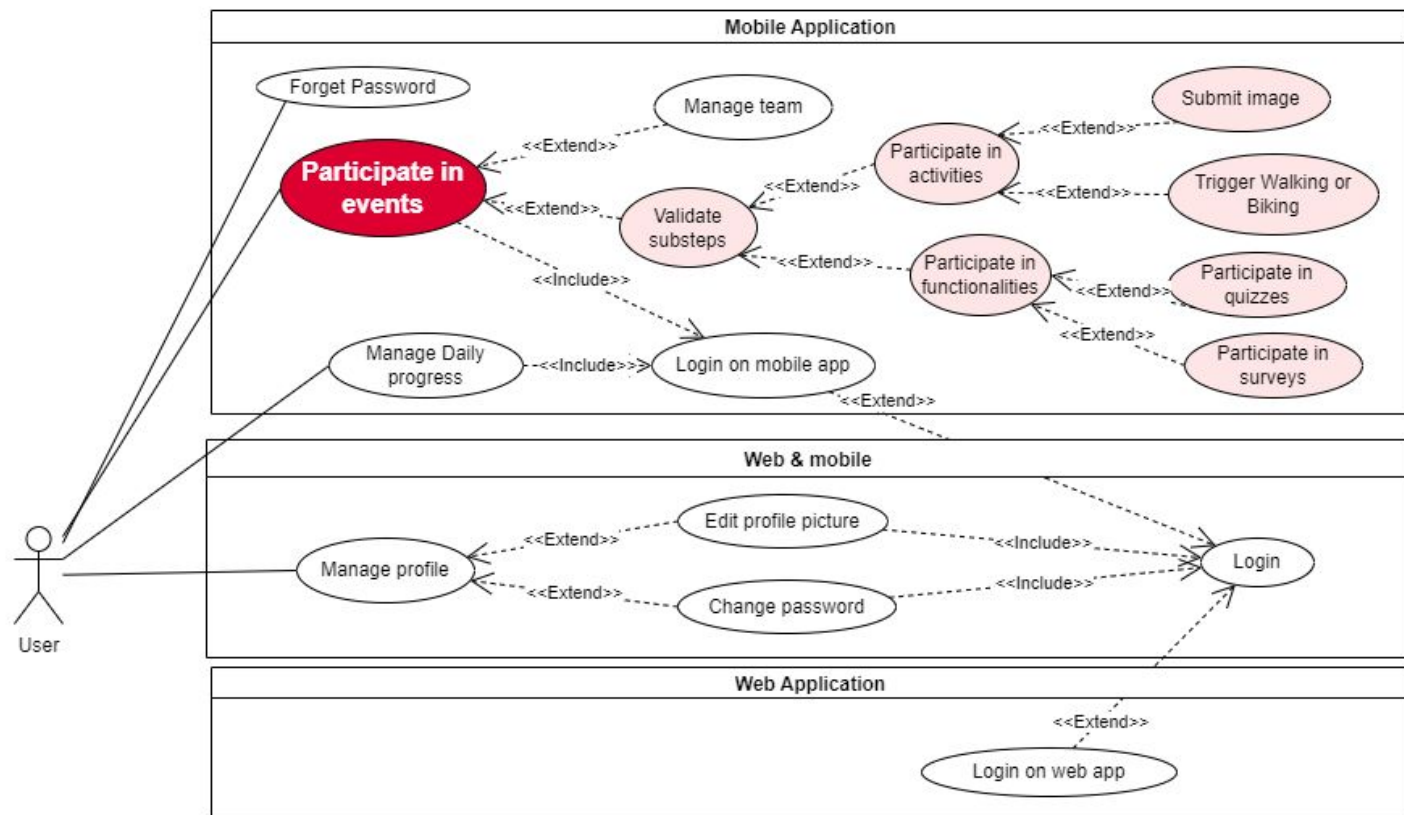
Sprint 2



Use case diagram detail



2nd Sprint : Green Challenge Application





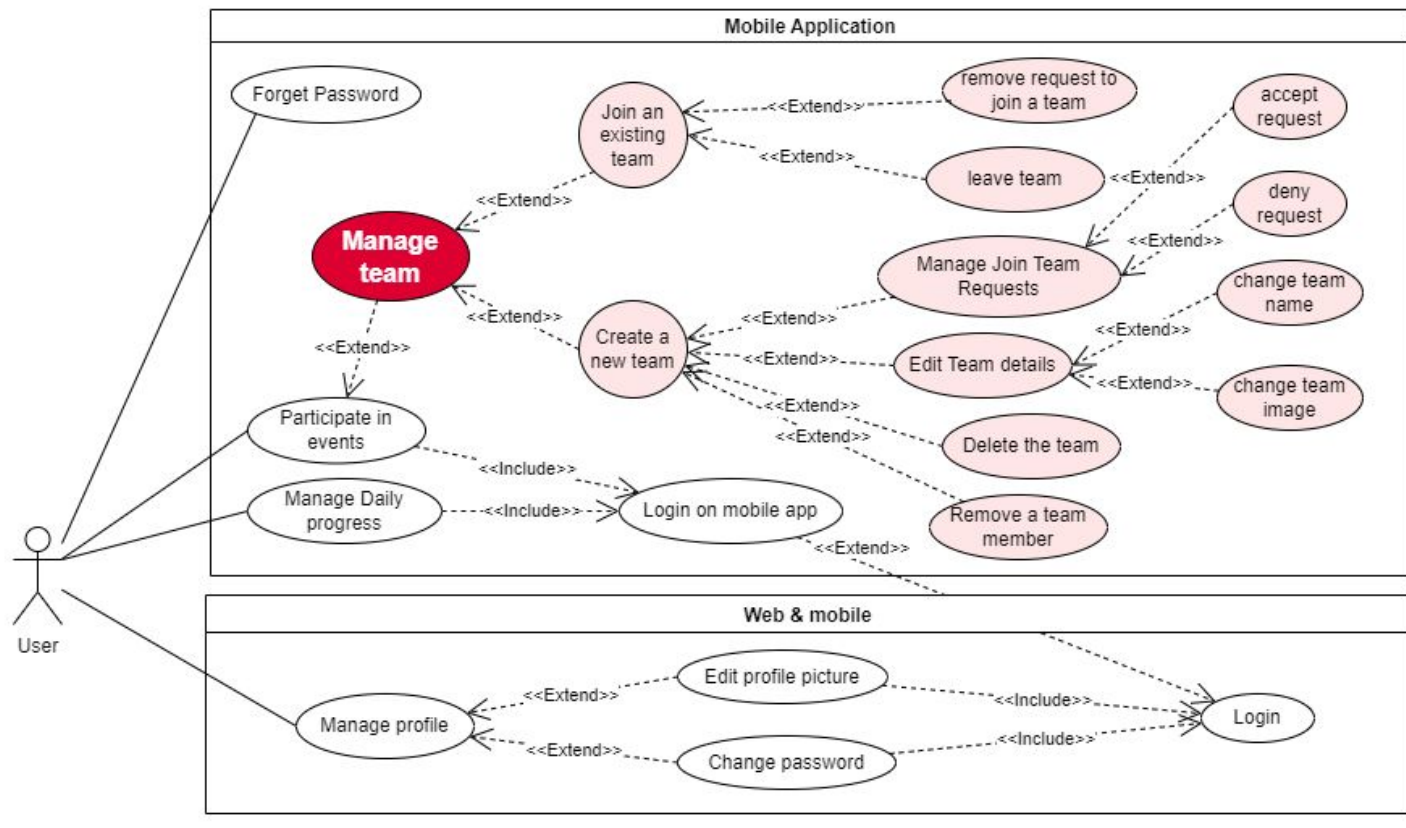
Sprint 2



Use case diagram detail



2nd Sprint : Green Challenge Application





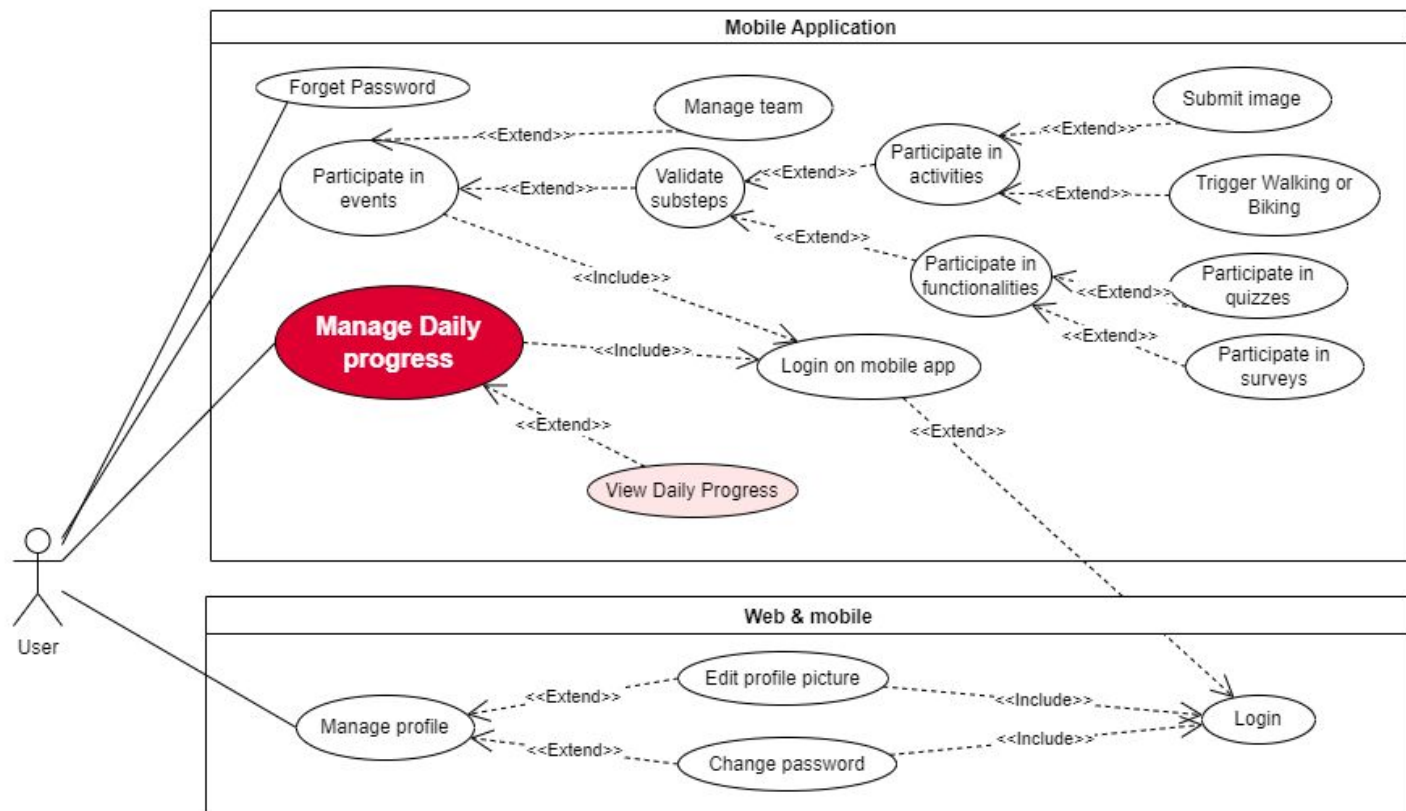
Sprint 2



Use case diagram detail



2nd Sprint : Green Challenge Application



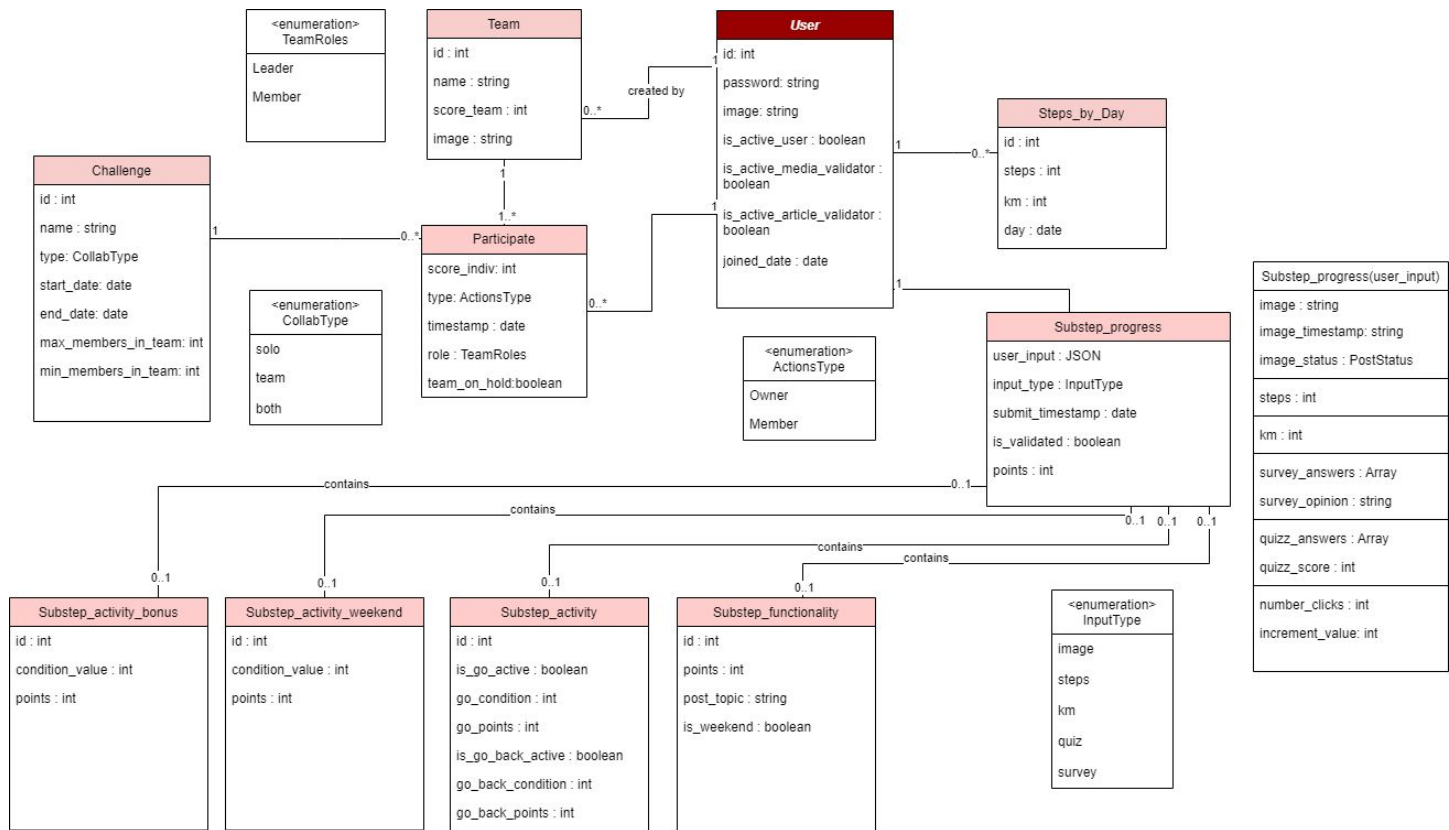


Sprint 2



Class Diagram

p63

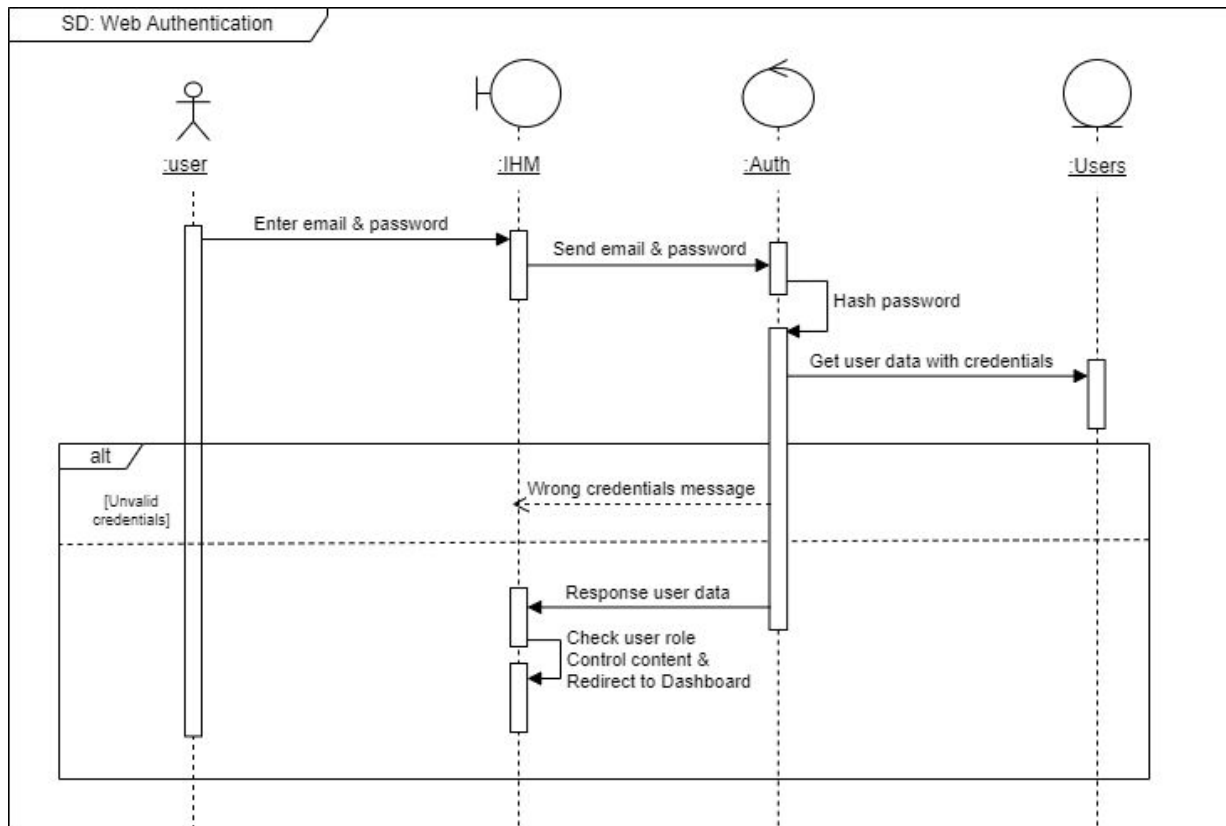




Authentication



Sequence Diagram p52





05

Sprint 3 : Secondary Features





Sprint 3

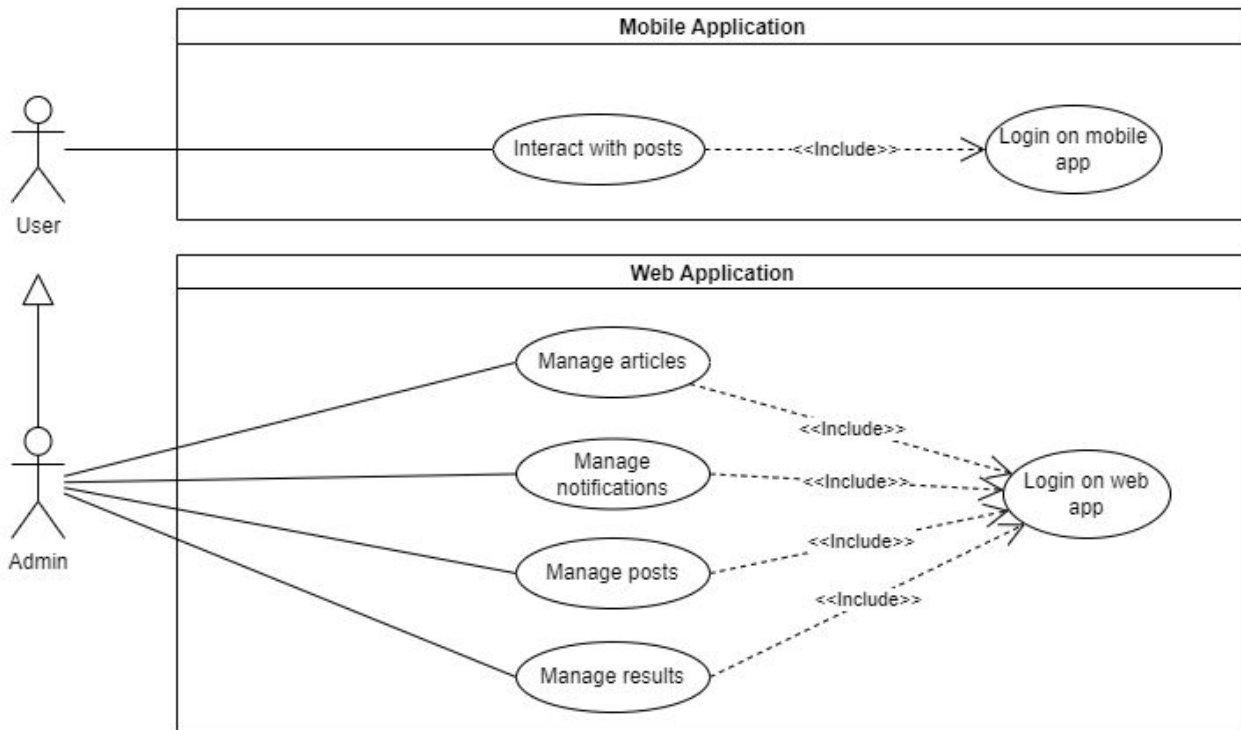


Use case diagram

p72



3rd Sprint : Green Challenge Application





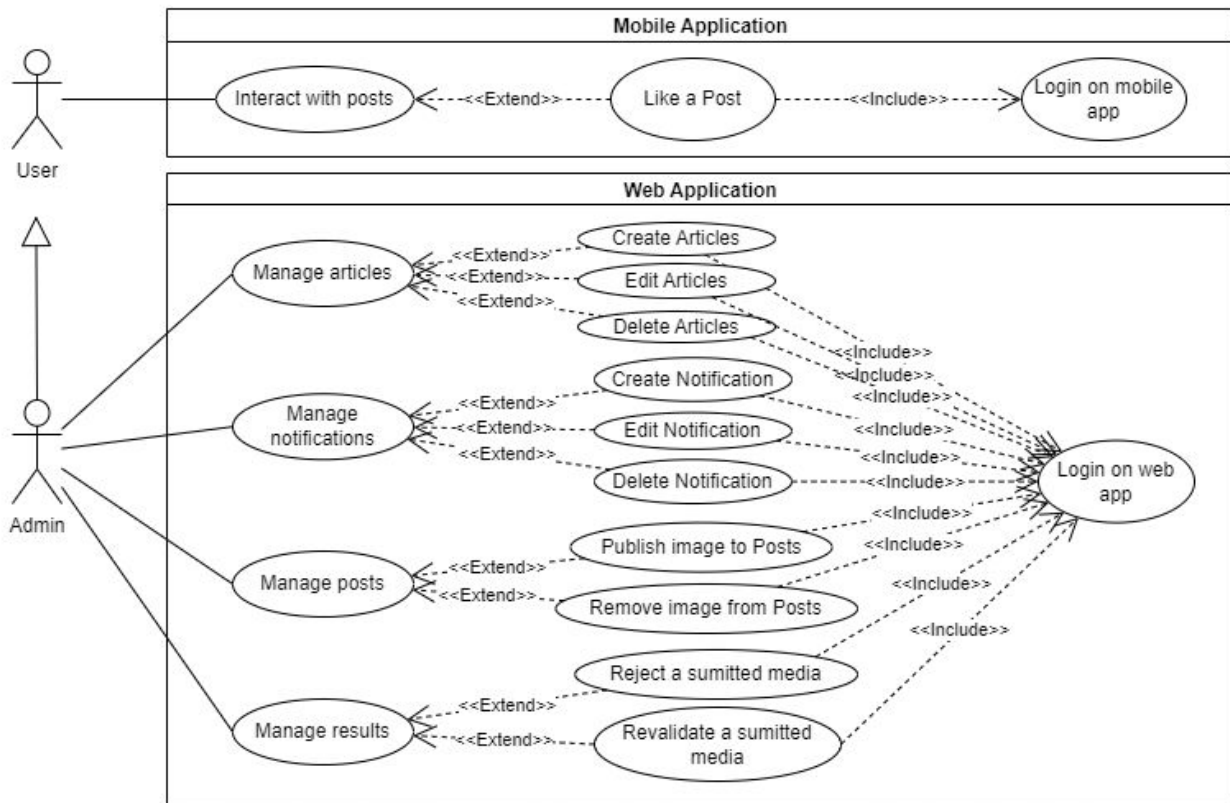
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application





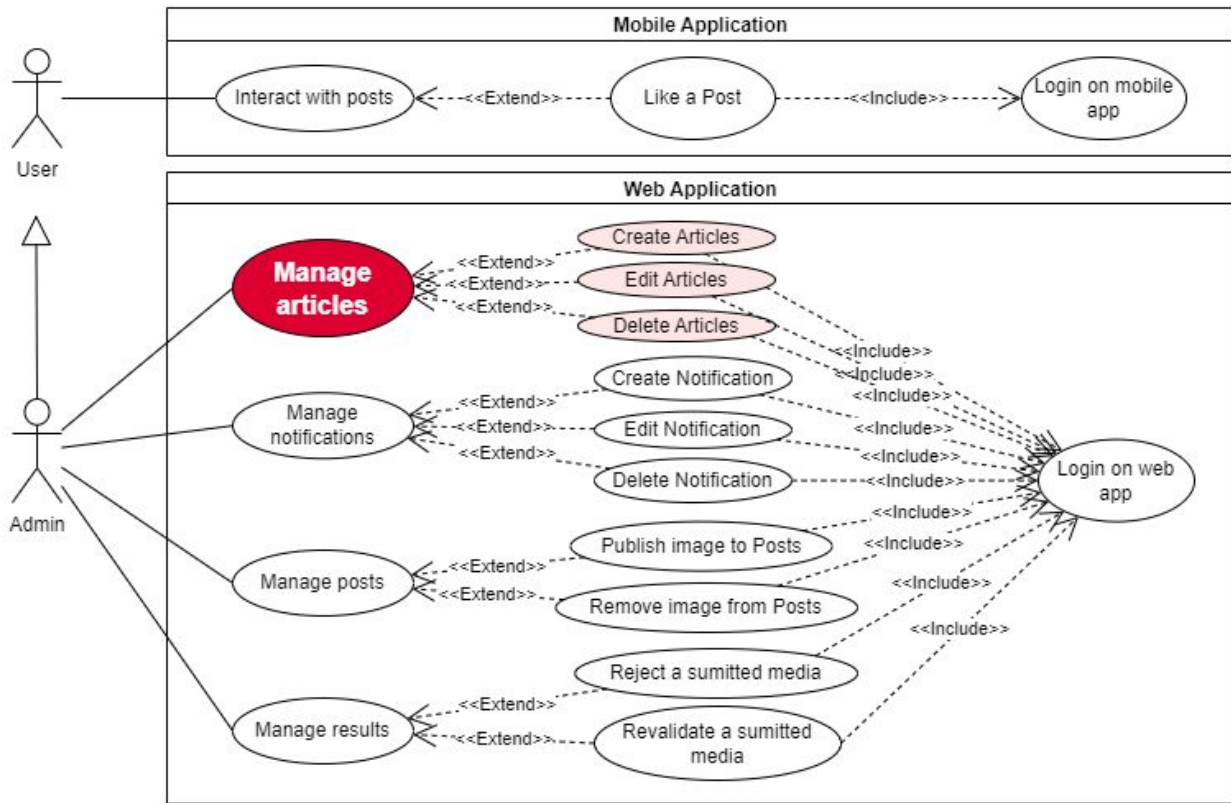
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application





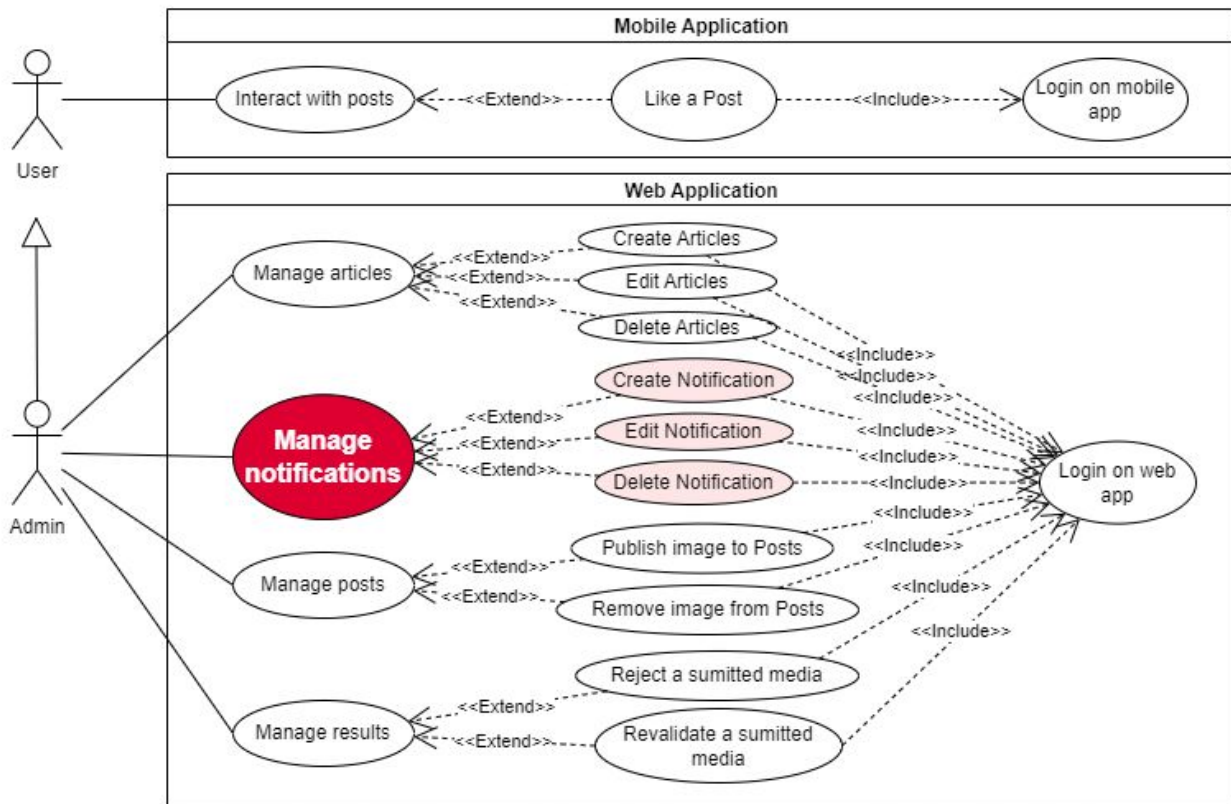
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application





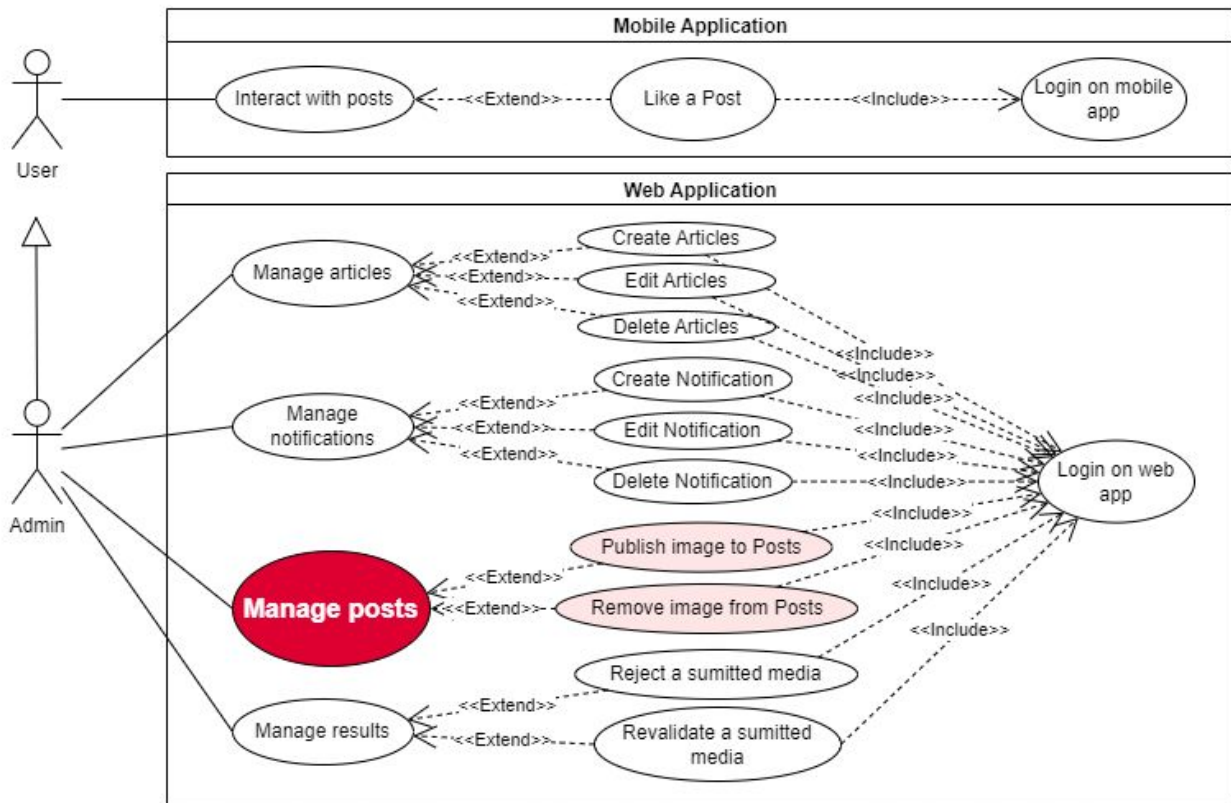
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application





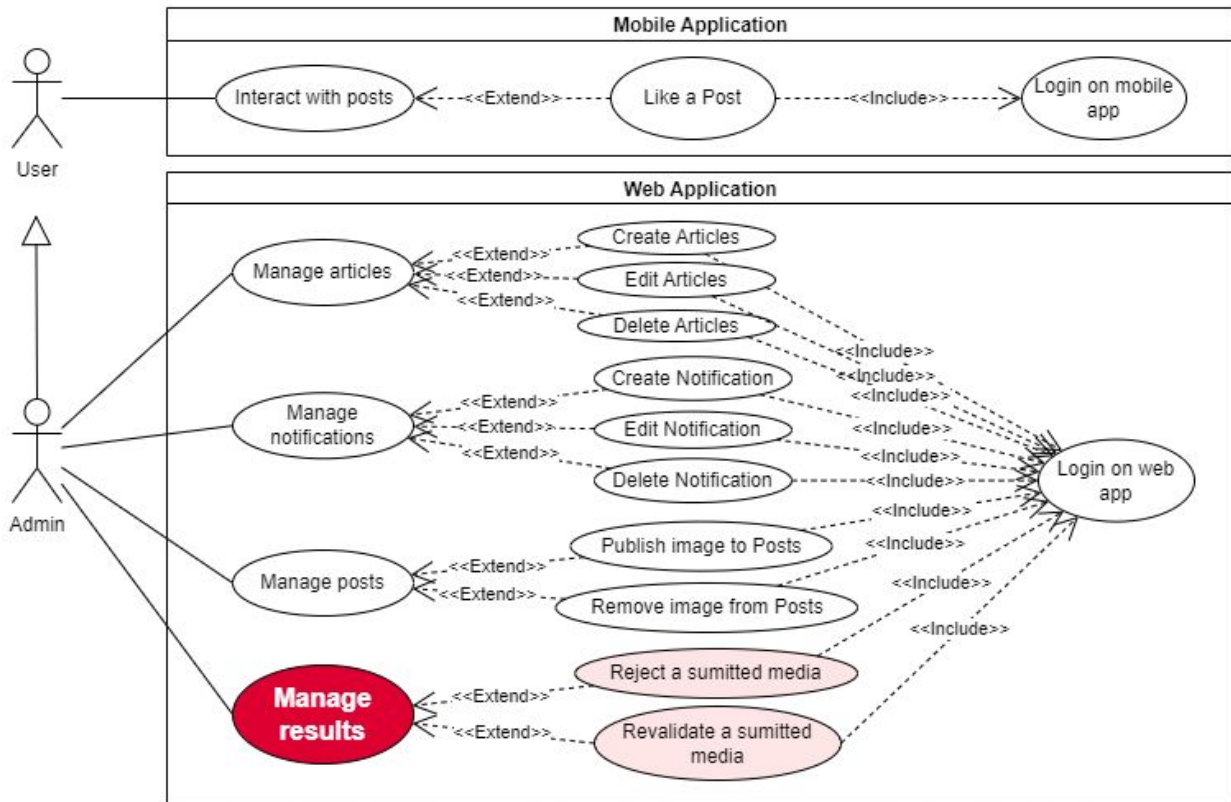
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application





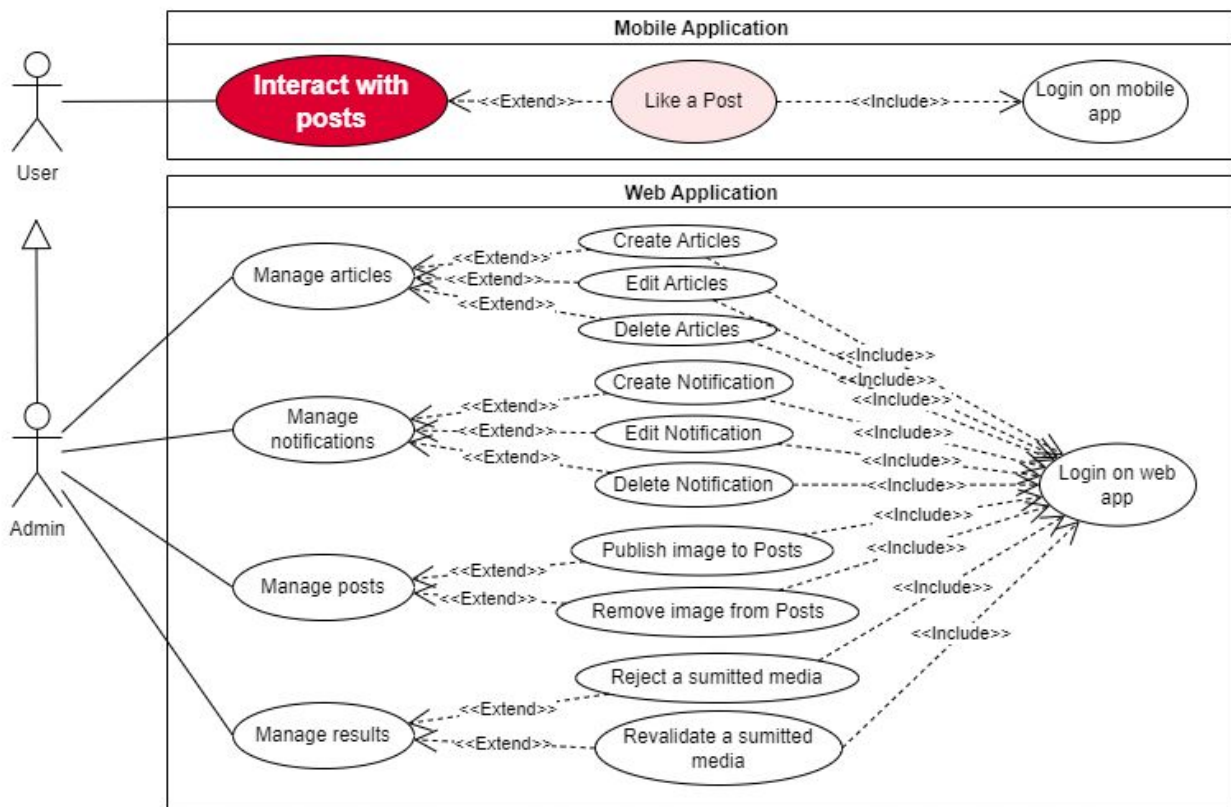
Sprint 3



Use case diagram detail



3rd Sprint : Green Challenge Application



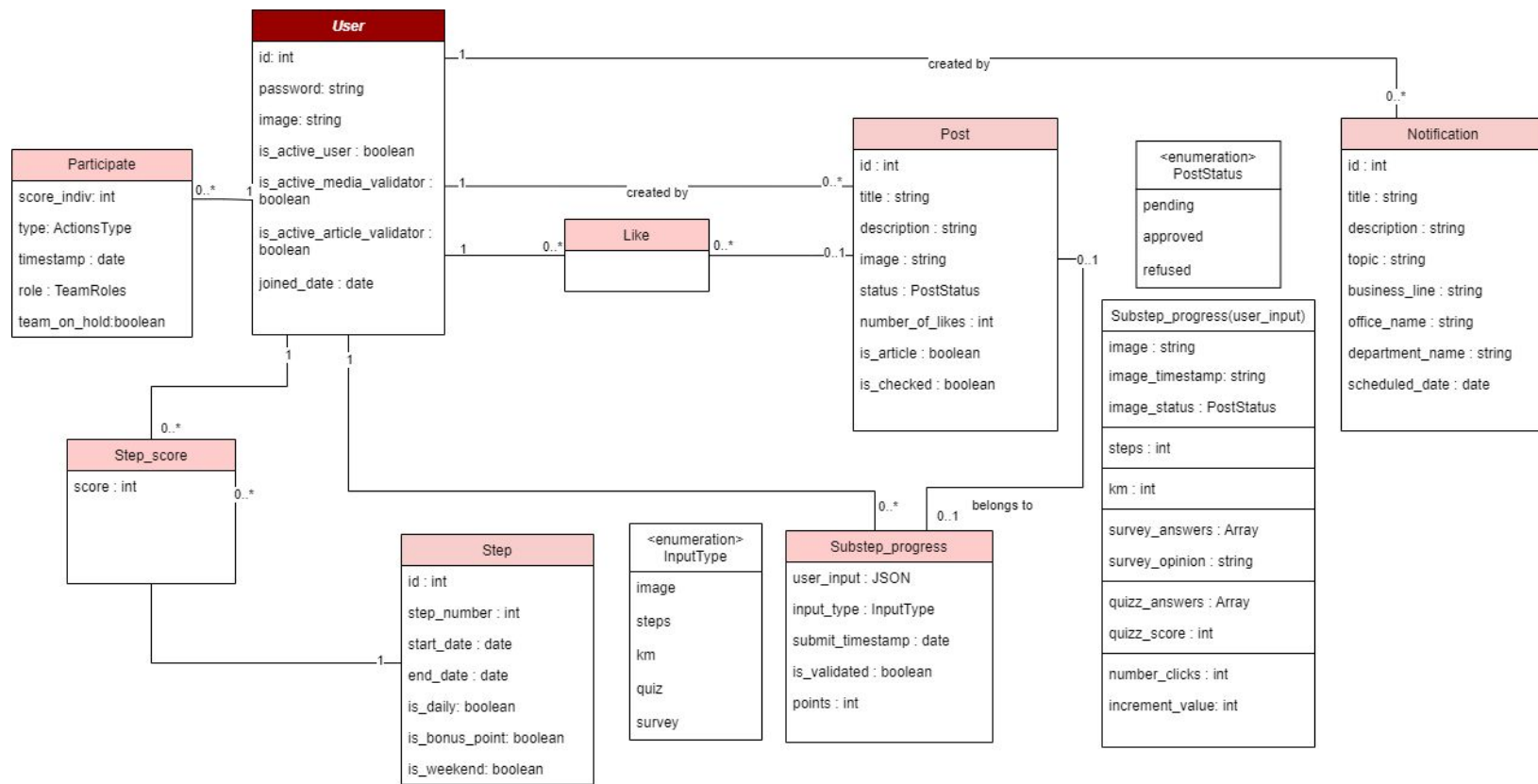


Sprint 3



Class Diagram

p81

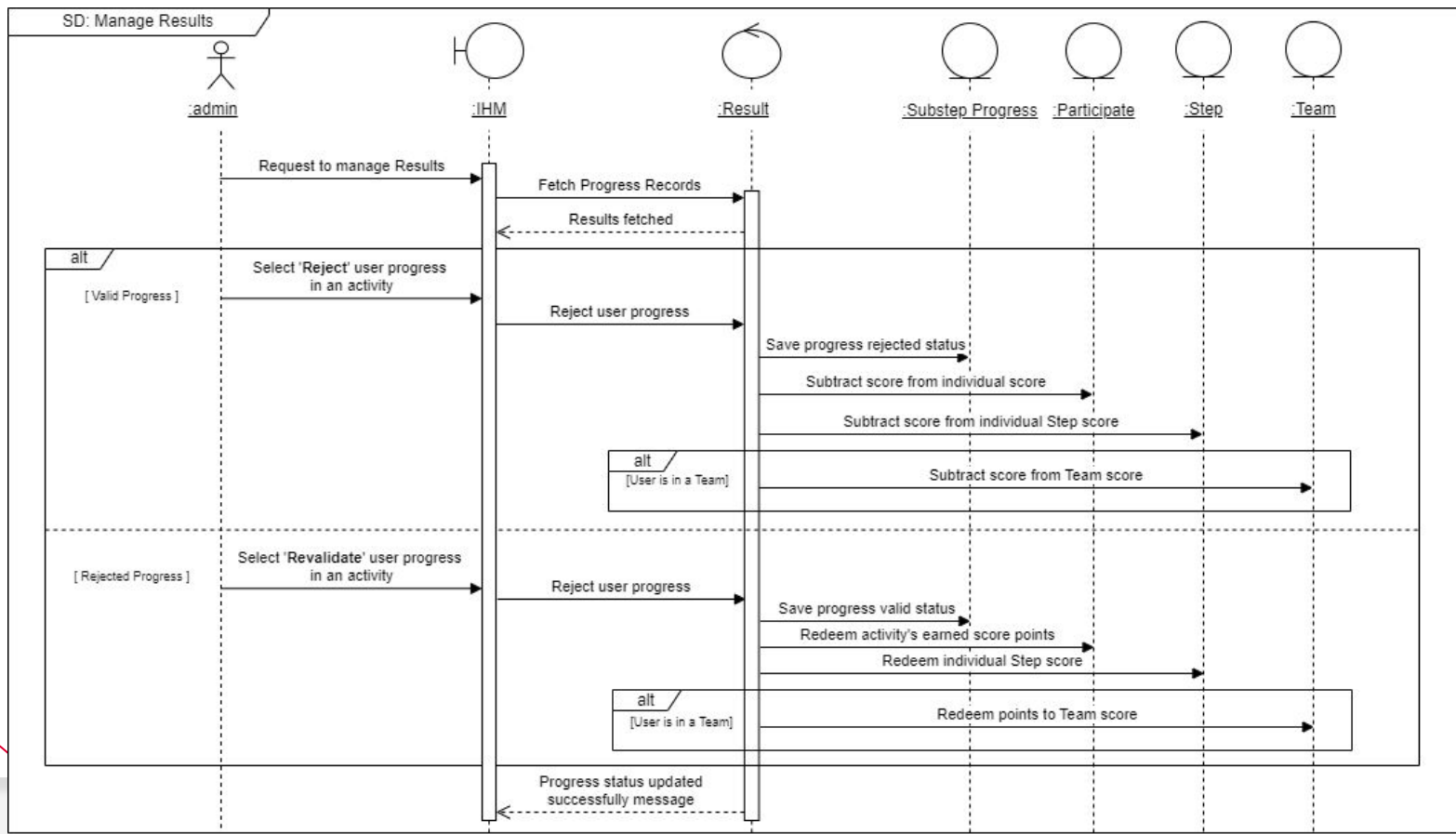


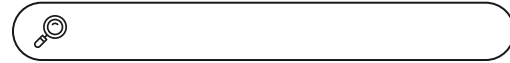


Event Results



Sequence Diagram p81





Web App

DEMO



Mobile App

DEMO

≡ Conclusion

 sustainable mobility



- By integrating **key functionalities** the apps are poised to drive engagement and facilitate efficient system oversight.
- Foundation for promoting **sustainable mobility** practices through a well-designed mobile application tailored to Linedata's objectives.
- Valuable technical **skills** in web and mobile development, alongside honing project management and communication abilities.



Perspective



- Advanced Step Tracking
- Improved web analytics
- Enhanced built-in social media : group chat, comments on posts ...
- Elevated User Experience : more offline functionalities, real-time collaborations



Thank You !

Do you have any questions?

